

Watsonx Orchestrate

Level 4 lab prep guide – March 2026 version



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Introduction

IBM lab environments are accessed directly from the course. Each course that includes a lab component has an activity within the course that allows you to easily access your own lab environments.

Each lab environment is specific to the product or use cases that are used in the course material. However, the basic steps for reserving and provisioning an environment are described in this document.

It can take some time for the lab environment to be ready for use. In some cases, it can be up to 90 minutes. In other cases, it might be less. After reserving your environment on TechZone, watch your mailbox. You will receive several emails about the status of your reservation (Provisioning / ready) or emails to activate IBM Cloud account.

The environments will be available to you for **5 days** from the time that you reserve it. Time can be extended one time following the IBM TechZone reservation policy.

About this lab

This preparation guide provides you with all the step-by-step instructions to get and configure the different environments or accounts that are required to run all the watsonx Orchestrate L4 lab activities.

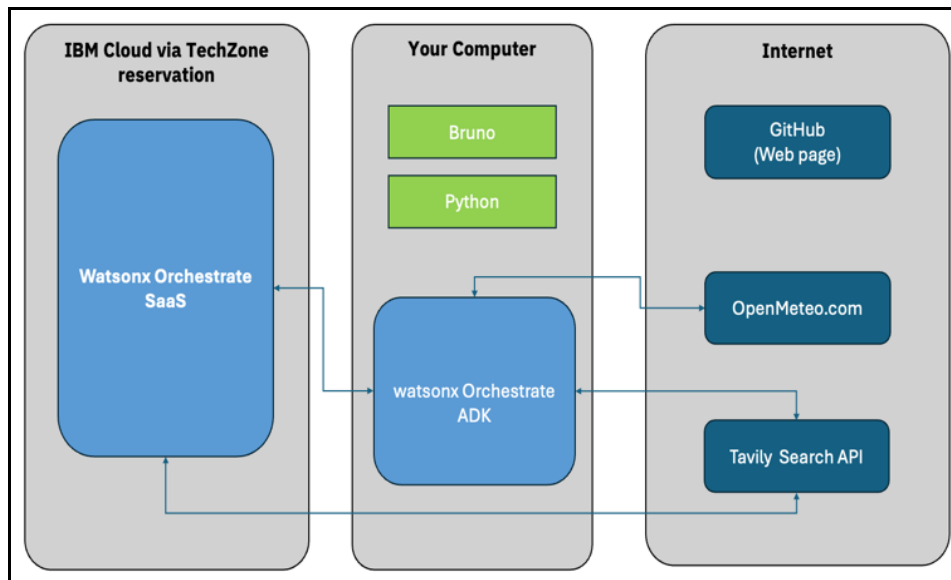
Pre-requisites

Before starting this lab, please make sure you have the following components installed on your local machine:

- Python 3.11: You can install Python via python.org (Recommended for most users) or using homebrew on Mac or Winget on MS™ Windows (Best for developers). AI assistants (MS™ Copilot /IBM Project Bob/Chat GPT...) can help you to proceed with this install.

High level architecture

This diagram gives you a high-level overview of the required environments to perform the watsonx Orchestrate Level 4 learning plan. You will need to reserve A watsonx Orchestrate SaaS hosted on IBM Cloud from IBM TechZone:



This guide will drive you through the different steps to provision these environments or create the 3rd party product accounts (Tavily in particular).

Lab credentials

During the watsonx Orchestrate L4 learning plan, you will navigate between different products, environments, accounts and services. This prep guide will help you to create this note to collect and reuse the different user/passwords, tokens and API keys you will need during the various lab activities.

This table summarizes the different items you will need to configure your lab environment. This guide will drive you through all the steps to collect them. Each item will be identified with its Item# to simplify the setup instruction in the different lab guides.

Environment	Credential	Value	Comments	Item #
Watsonx Orchestrate SaaS	Account name		The ibm cloud account name you have been onboarded onto.	Item 1
	Service URL		The service url of your watsonx Orchestrate SaaS instance	Item 2
	API KEY		The API key you have generated to access your watsonx Orchestrate SaaS tenant	Item 3
Tavily Web Search (Used to create an agentic tool)	Command line	<code>npx -y tavily-mcp@0.1.3</code>		Item 4
	TAVILIY_API_KEY			Item 5
Localhost user	username	wxo.archer@ibm.com		Item 6
	password	watsonx		Item 7

Create an account to request your environment

IBM employees can use their IBM credentials to access IBM Technology Zone or IBM Cloud.

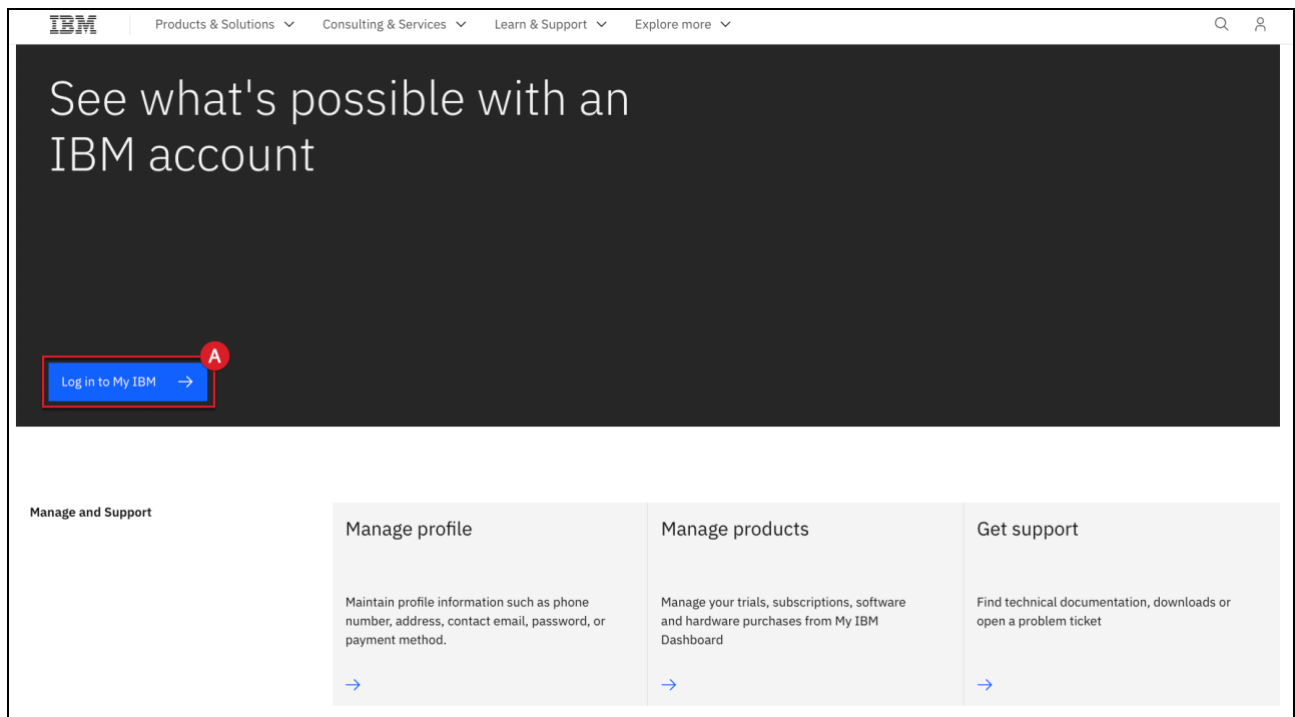
Business Partners must use an IBMid for IBM Technology Zone. If you do not have an IBMid, you can create one by using steps in this document. There is no cost to creating an IBMid.

You might also require an IBM Cloud account for some lab environments.

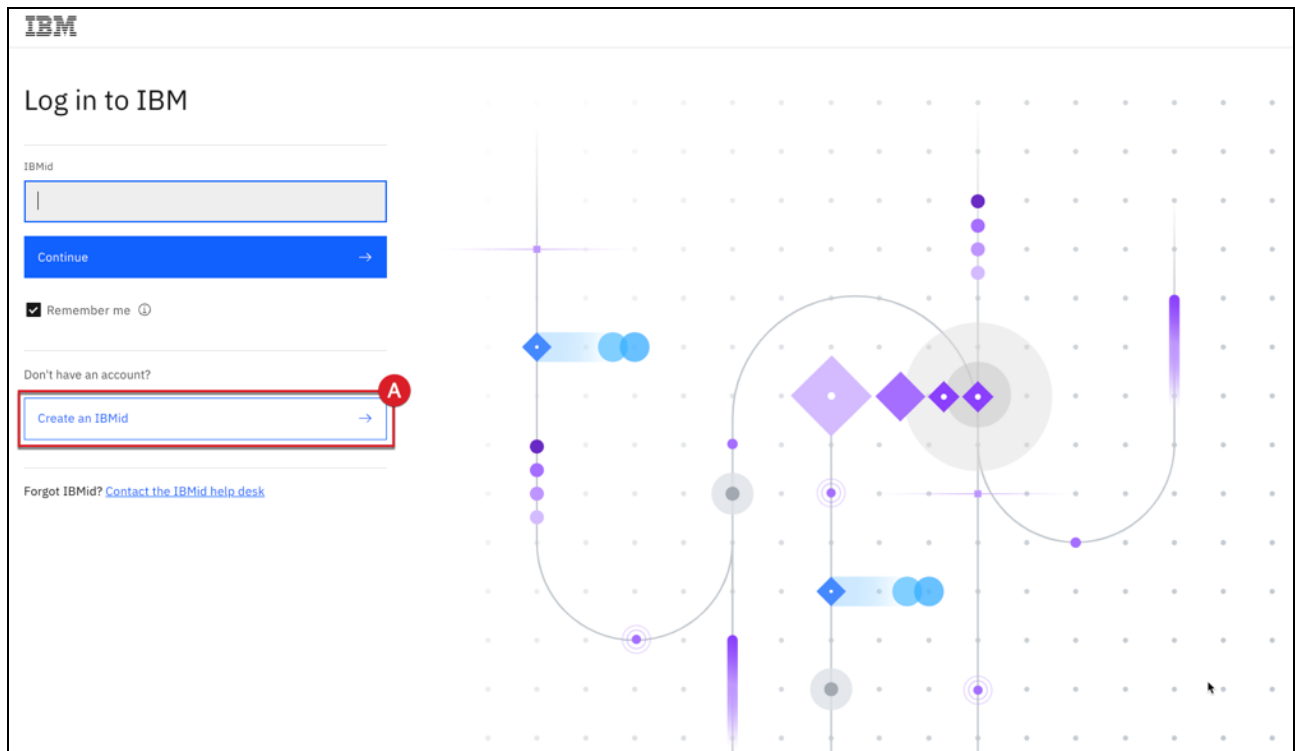
Additionally, you might need to configure an OpenVPN connection to be able to access the lab environment.

Creating an IBMid

1. In your web browser, go to the IBM account page (<https://www.ibm.com/account/us-en/>).
1. Click **Log in to My IBM** (A).



2. The Log in to IBM page is displayed. Click **Create an IBMid** (A).



3. On the **Create an IBMid** page, enter the following information:

- Your email address (A)
- A password for your IBMid account (B)
- First name (C)
- Last name (D)
- Your country or region of residence (E) and state or province (F)
- If you select that you are not a student (G), then enter the name of your company (G)
- Click **Next** (I) to continue

The screenshot shows the 'Create an IBMid' page. On the left, a 'Welcome to IBM' section encourages creating an account. The main form on the right is titled 'Create an IBMid' and includes a link for existing users. The form is divided into sections: 'Account information' and 'Are you a student?'. The 'Account information' section contains fields for Business email (A), Password (B), First name (C), Last name (D), Country or region of residence (E), and State or province (F). The 'Are you a student?' section has radio buttons for 'Yes' and 'No' (G). Below this is a 'Company' field (H) and a 'Next' button (I). Red boxes and letters A-I are overlaid on the form to indicate the required input fields.

Welcome to IBM

Create an account to access trials, demos and services.

Create an IBMid

Already have an IBM account? [Log in](#)

Account information

Business email (A) itzuser@email.com (B)

We'll send you a 7 digit code that you'll use to verify your email in step 2.

First name (C) ITZuser (D)

Last name Techzone

Country or region of residence (E) United States of America (F)

State or province Texas

Are you a student? (G)

☐ Yes ☒ No

Company (H) IBM

Next (I)

4. You are then prompted to verify your email and complete your account creation. Enter the code that was emailed to you in **Verification token** (A), and then click **Create account** (B).

Welcome to IBM

Create an account to access trials, demos and services.

Create an IBMid

Already have an IBM account? [Log in](#)

Account information

Verify email

We emailed a 7 digit code to [itzuser@email.com](#). This code will expire in 30 minutes.

Verification token

5548663

Didn't receive the email? Check your spam filter for an email from [ibmacct@iam.ibm.com](#).

You can resend the verification token in 43 seconds

I'd like IBM to use my contact details to keep me informed about products, services, and offers. More information on how IBM uses data and ways to [opt out](#) can be found in the [IBM Privacy Statement](#).

☒ Email

I accept the product [Terms and Conditions](#).

Create account

5. The **About your IBMid Account** page is shown. Click **Proceed** (A).

IBM

Welcome to IBM

Create an account to access trials, demos and services.

About your IBMid Account

By creating, or using, an IBMid to access an IBM Web site or other service, you acknowledge that you have read, understand, and agree to be bound by the below terms, in addition to the terms in the IBM Terms of Use, the IBM Privacy Agreement and the "About your IBMid Account Privacy" section of this notice provide details on how IBM protects your personal data.

Most IBMid accounts are individual IDs, meaning you manage your own account information and access your account through the IBMid login service (i.e. IBMid is the authenticating agency).

If your account is created using an email address containing a domain owned by an organization which you are employed by, contracted to, or volunteer for, your organization can:

- inquire about the status of your ID
- request your account settings (including your personal information)
- at its option, convert it to an enterprise ID.

If your account was created as (or is converted to) an enterprise ID, your organization manages your account information. It is also the authenticating agency for your ID, meaning your access is controlled by your organization's login service.

About your IBMid Account Privacy

What data does IBM collect?

Why IBM needs your data

How your data is obtained

How IBM uses your data

How IBM protects your data

How long we keep your data

Your rights

I acknowledge that I understand how IBM is using my Basic Personal Data and (if applicable) how my organization may become the authenticating agency for my IBMid account. I certify that my age is at least 18 years old.

Cancel

Proceed

- After a few minutes, you will receive an email confirming that your IBMid has been activated.

Note: The first time that you log in using your IBMid you will be sent another code to verify your log in.

Creating an IBM Cloud Account (optional)

If you already have an IBM Cloud account, you can skip this task.

- Go to IBM Cloud (<https://cloud.ibm.com/registration>).
- Enter your email address (A) and a password (B), and then click **Next** (C).

The screenshot shows the 'Create an IBM Cloud account' page. The 'Account information' section is active, showing fields for 'Email' (labeled A) and 'Password' (labeled B). Below these fields is a 'Next' button (labeled C) with a downward arrow. To the right of the form, there are three benefits listed: 'Unlock the full IBM Cloud Catalog', '200 USD Cloud Credit for free', and 'Get started for free'. At the bottom of the form, there are links for 'Verify email', 'Personal information', and 'Account notice'. A 'Create account' button is at the very bottom.

After you complete the registration, you receive an email to activate your account. This can take a few hours to process. After you receive the confirmation email, follow the instructions provided in the email to activate your account.

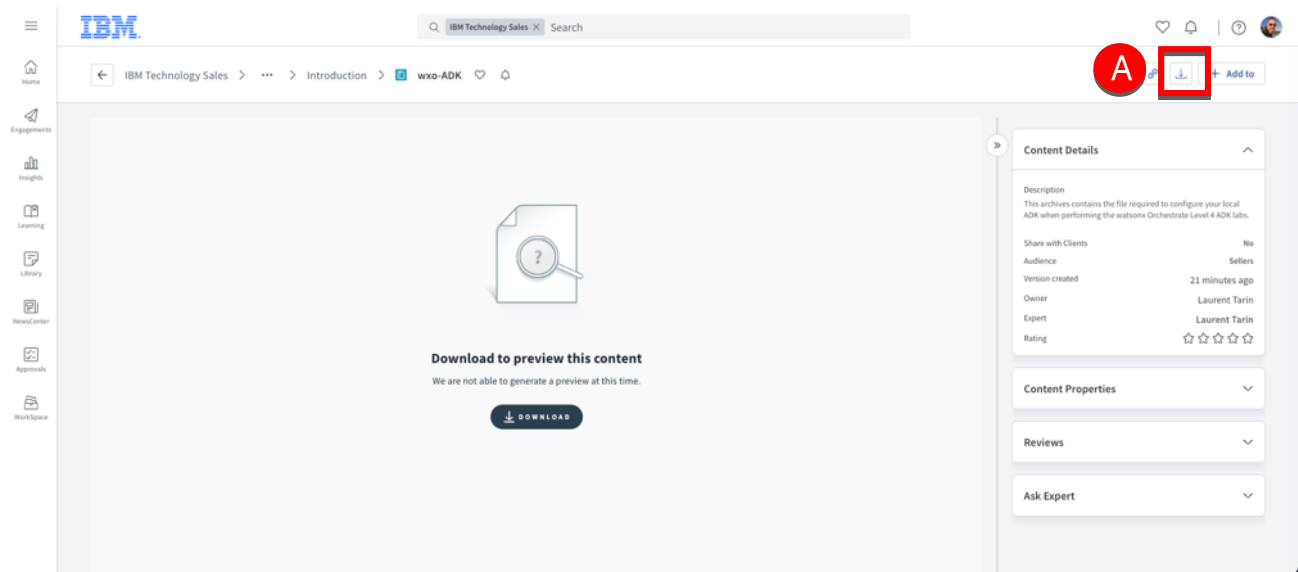
Prepare an ADK working directory

Download the required files

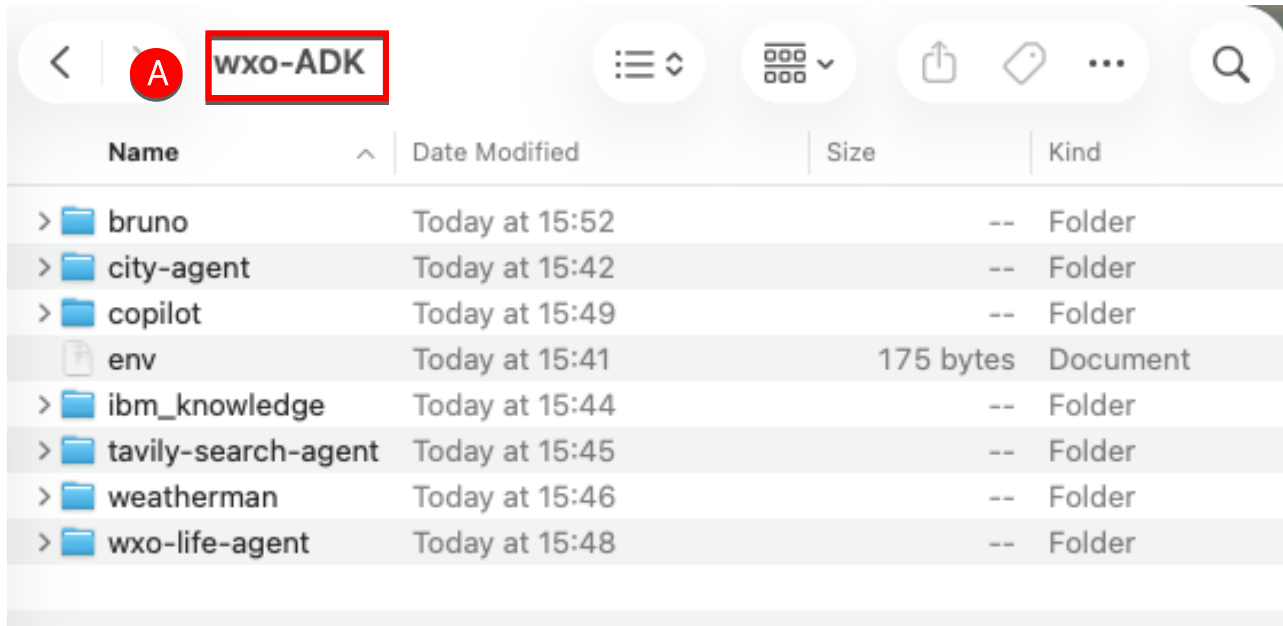
In this step, you will get download all the files that are required to run the various ADK labs on your own computer.

It is strongly recommended to create the virtual environment directly from within this directory once un-archived on your computer.

1. Visit <https://ibm.seismic.com/Link/Content/DCqWGcVp2V34qGqG829PQFpf99cV>
2. Click the download link (A).



3. Extract the archive file on your computer. In this lab guide, the file has been extracted under the home directory (~).
4. Open your local **wxo-ADK** folder (A).



This folder contains:

- The **bruno** folder containing the workspace configuration file.
- The **city-agent** file used to test your ADK environment.
- An empty **copilot** folder to store the copilot generated agent files.
- An **env** file you will edit to specify your watsonx Orchestrate server credentials
- 4 folders containing the required files to run the ADK labs.

Reserve and prepare your SaaS environment

In this section you will reserve and configure two different environments (described in the architecture diagram) that you will use through the different labs in the watsonx Orchestrate L4.

Watsonx Orchestrate SaaS

During this L4 you will need a wxO SaaS trial to perform most of the labs contained in the new watsonx Orchestrate learning plan (September version). Follow these instructions to reserve and configure this SaaS environment.

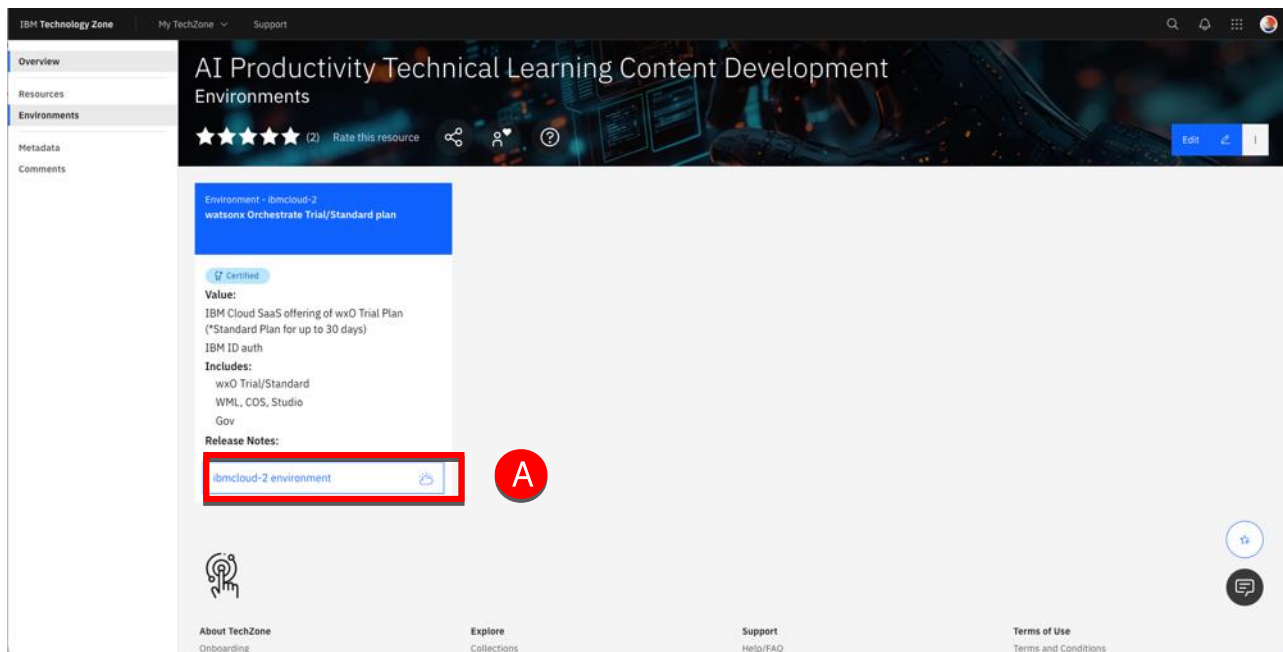
Reserve the environment

The environment can be reserved from IBM TechZone.

1. Click the following link to access the TechZone collection.

<https://techzone.ibm.com/collection/ai-productivity-technical-learning-content-development/environments>

2. Click **Reserve it** in the **watsonx Orchestrate Trial / Standard plan** tile (A).



- Click **Request an environment** (A).

IBM Technology Zone My TechZone Support

Create a reservation

AI Productivity Technical Learning Content Development

View collection

Select an environment/infrastructure Select a reservation type Fill out your reservation Complete

Select your reservation type

environment reservation options:

☒ Request an environment (A)

☐ Schedule a workshop

Request multiple environments:

Leverage workshop manager to request multiple environments for your upcoming hands-on workshop and manage them all in one place. All workshop requests must be submitted 7 days before the intended workshop start date/time.

I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

Cookie Preferences

- Select **Education** (A).

IBM Technology Zone My TechZone Support

Create a reservation

AI Productivity Technical Learning Content Development

View collection

Select an environment/infrastructure Select a reservation type Fill out your reservation Complete

Name

watsonx Orchestrate Trial/Standard plan

Name this reservation. This will help identify it in your reservation list.

Purpose

Education (A)

Gaining experience with specific technology, product, or solution.

Demo

Deliver a client specific demonstration based on discovery with the client and aligns to the identified architecture. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Lab Development

Build and configure environments to support the creation and testing of labs for the TechXchange event. Intended for lab developers during the development phase before final student delivery.

Pilot

Rapid co-creation build that proves IBM technologies can deliver business value to clients' and users. Serves as a foundation to build a production solution. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Test

Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can NOT be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Preferred Geography

Choose a preferred geography

I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

Cookie Preferences

5. Type a Purpose description (A).

IBM Technology Zone | My TechZone | Support

Purpose

Demo
Deliver a client specific demonstration based on discovery with the client and aligns to the identified architecture. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Education
Gaining experience with specific technology, product, or solution.

Lab Development
Build and configure environments to support the creation and testing of labs for the TechXchange event. Intended for lab developers during the development phase before final student delivery.

Pilot
Rapid co-creation build that proves IBM technologies can deliver business value to clients' end users. Serves as a foundation to build a production solution. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Test
Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can **NOT** be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Sales Opportunity number

Enter an opportunity number

Providing an [IBM Sales Cloud opportunity number](#), or a [Project Work ID](#) will allow you to extend your reservation date.

Purpose description

wxO L4 labs

What are you doing? Why do you need this? What are you trying to accomplish?

Preferred Geography

Choose a preferred geography

☐ I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

6. Select AMERICAS – us south region (A) in the Preferred Geography menu.

IBM Technology Zone | My TechZone | Support

Purpose

Demo
Deliver a client specific demonstration based on discovery with the client and aligns to the identified architecture. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Education
Gaining experience with specific technology, product, or solution.

Lab Development
Build and configure environments to support the creation and testing of labs for the TechXchange event. Intended for lab developers during the development phase before final student delivery.

Pilot
Rapid co-creation build that proves IBM technologies can deliver business value to clients' end users. Serves as a foundation to build a production solution. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Test
Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can **NOT** be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Sales Opportunity number

Enter an opportunity number

Providing an [IBM Sales Cloud opportunity number](#), or a [Project Work ID](#) will allow you to extend your reservation date.

Purpose description

wxO L4 labs

What are you doing? Why do you need this? What are you trying to accomplish?

Preferred Geography

✓ R2-watsonx - AMERICAS - us-south region - dal10 datacenter

R2-watsonx - AP - jp-tok region - tok02 datacenter

☐ I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

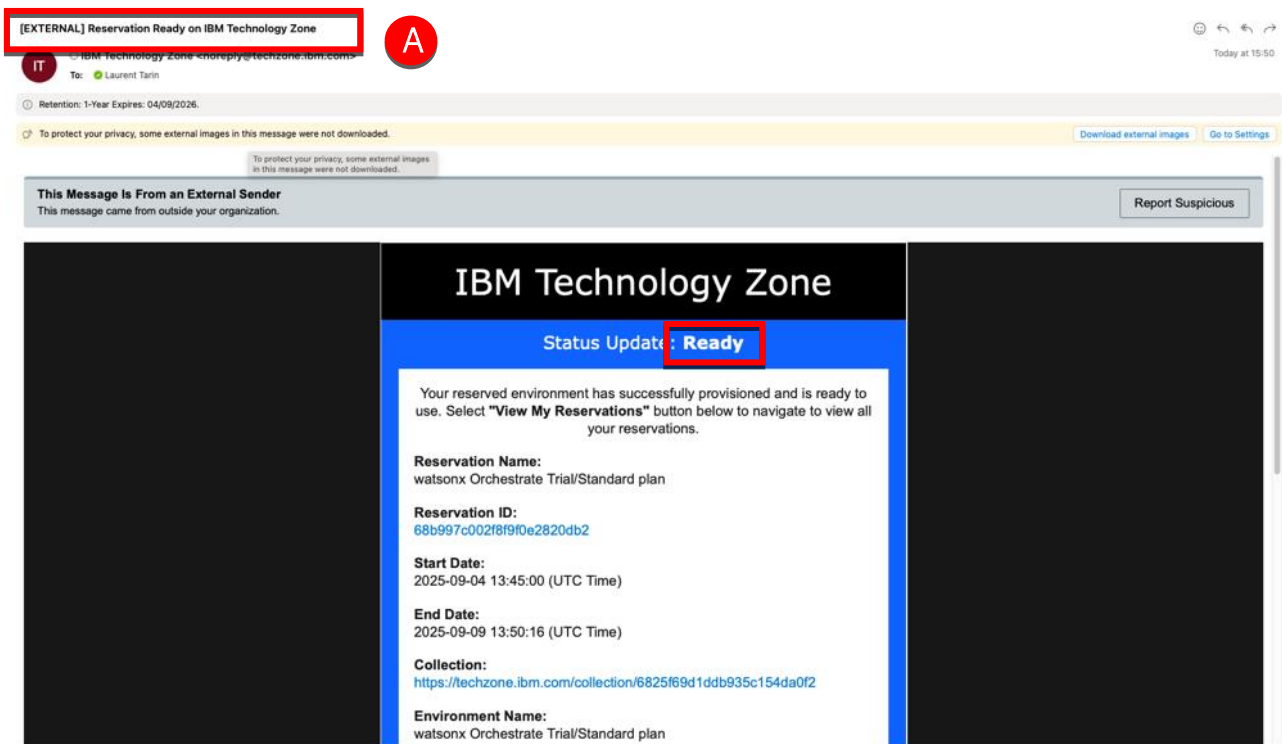
7. Read and **Agree the IBM Technology Zone** conditions (A) then click **Submit** (B).

The screenshot shows the 'Create a reservation' form in the IBM Technology Zone. The form includes sections for Purpose (Demo, Education, Lab Development, Pilot), Sales Opportunity number, Purpose description (wxO L4 labs), and Preferred Geography (itz-watsonx - AMERICAS - us-south region - dal10 datacenter). A red box labeled 'A' highlights the 'I agree to IBM Technology Zone's Terms & Conditions and End User Security Policies' checkbox, which is checked. Another red box labeled 'B' highlights the 'Submit' button.

8. Check that your reservation has been created (A) then wait for the “reservation is ready” email.

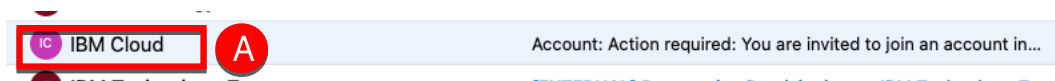
The screenshot shows the 'Thank you' page after a reservation has been created. The page displays the message: 'Your ibmcloud-2 reservation has been created for watsonx Orchestrate Trial/Standard plan. You will receive an email at laurent.tarin@fr.ibm.com with information about your reservation once it is ready. You can manage your reservations on the My reservations page.' A red box labeled 'A' highlights a notification banner in the top right corner that reads: 'Reservation: Reservation "watsonx Orchestrate Trial/Standard plan" created Sep 4, 2025 3:44 PM (+02:00)'. Below the message, there is a 'My reservations' button.

9. Wait for the READY confirmation email from IBM Technology zone (A):



Join the IBM Account

Your watsonx Orchestrator instance is hosted on an IBM Cloud account. You will first need to join this account. If you have received another email from IBM Cloud (A) inviting you to join an account, open this email.



- The screenshot shows an email interface. At the top left, there's a header bar with a purple circle containing 'IC' and the text 'IBM Cloud <no-reply@cloud.ibm.com>' and 'To: Laurent Tarin'. On the right, it says 'Today at 14:30'. Below the header, there's a yellow banner stating 'Retention: 1-Year Expires: 25/08/2026.' and a note 'To protect your privacy, some external images in this message were not downloaded.' with links for 'Download external images' and 'Go to Settings'. The main body of the email has a dark header with 'IBM Cloud' and the 'IBM Cloud Logo'. Below this, the word 'Account' is prominently displayed. The text reads: 'Action required: You are invited to join an account in IBM Cloud'. It lists details: 'Source ID: 1756125020166', 'Type: Account', and 'Update Time: 25 Aug 2025, 12:30 PM UTC'. A link 'View notification (In subscribed accounts):' is provided. The greeting is 'Hi laurent.tarin@fr.ibm.com,'. Two red circles with white letters 'A' and 'B' highlight specific parts: Circle 'A' points to the account name 'account: itz-saas-332 (Account ID: 6077217005251500d000c0002a00f78e)' which is enclosed in a red box. Circle 'B' points to a blue 'Join now.' link, also enclosed in a red box. The footer includes 'Thank you, IBM Cloud' and 'In case of any issues, please contact IBM Cloud support.' along with a copyright notice '© Copyright IBM Corporation 2014, 2025.'

-
- IBM Cloud
- Catalog Cost estimator Docs
- ## Join IBM Cloud
- You're almost there. Complete your IBM Cloud account details so that you can get started.
- Account information

laurent.tarin@fr.ibm.com

Personal information

Laurent Tarin

United States

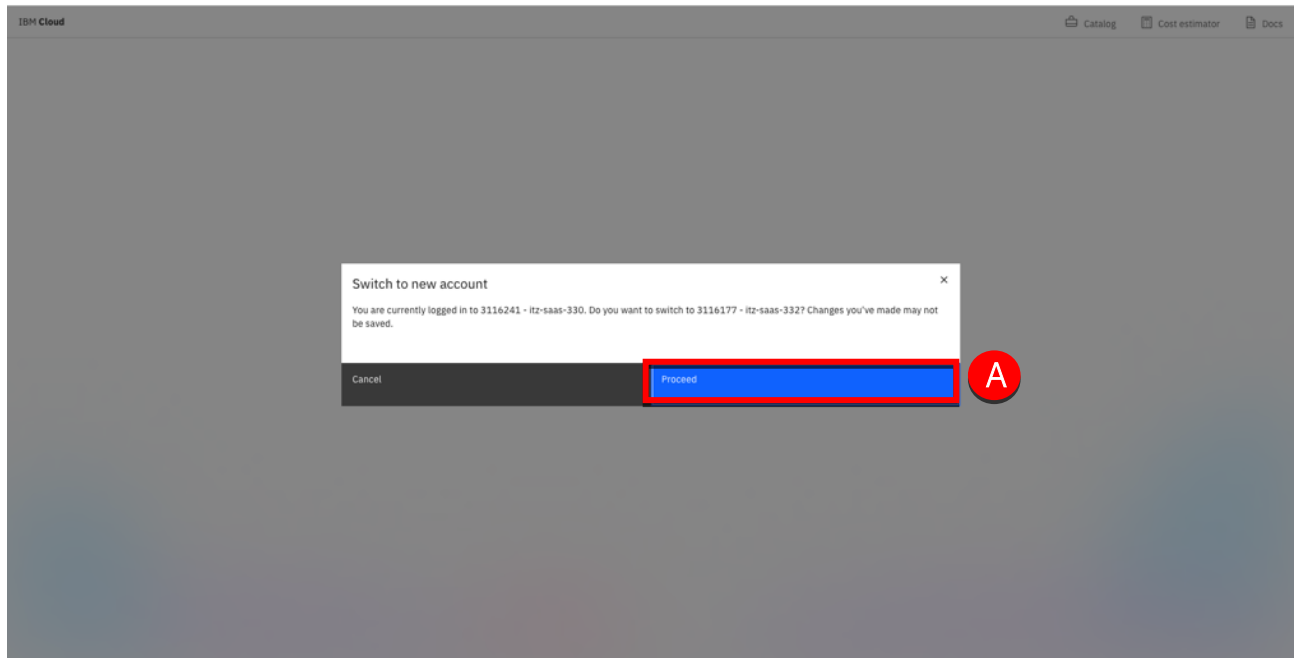
Account notice

By providing my contact details, I acknowledge that IBM will keep me informed about products, services, and offers. More information on how IBM uses data can be found in the [IBM Privacy Statement](#). California residents, review the [California Supplemental Privacy Statement](#).

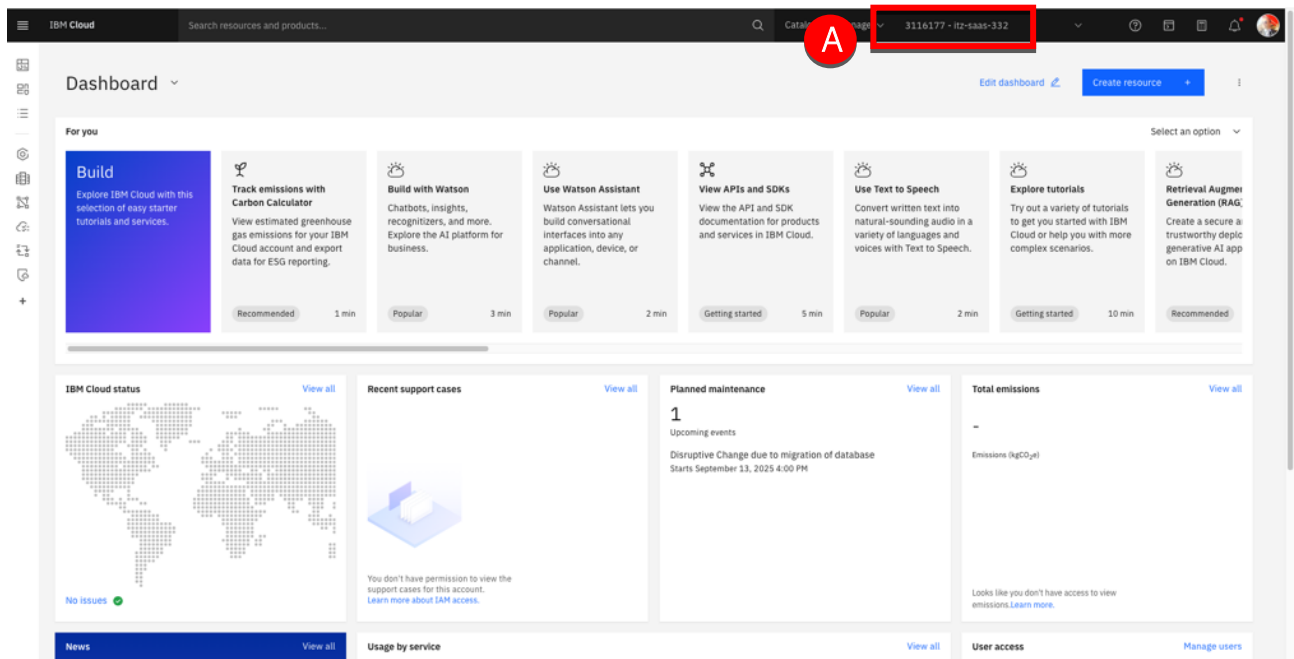
☒ I accept the product [Terms and Conditions](#) of this registration form.

Join account
- Unlock the full IBM Cloud Catalog
- Get started with everything you need to take your projects to the next level. Including always-free products like IBM Watson® APIs. They never expire and you can't be charged for them—ever.
- Cookie Preferences

5. Click **Proceed** to switch to the new account on IBM Cloud (A). Note, you will not see this message if this is your only IBM Cloud account.



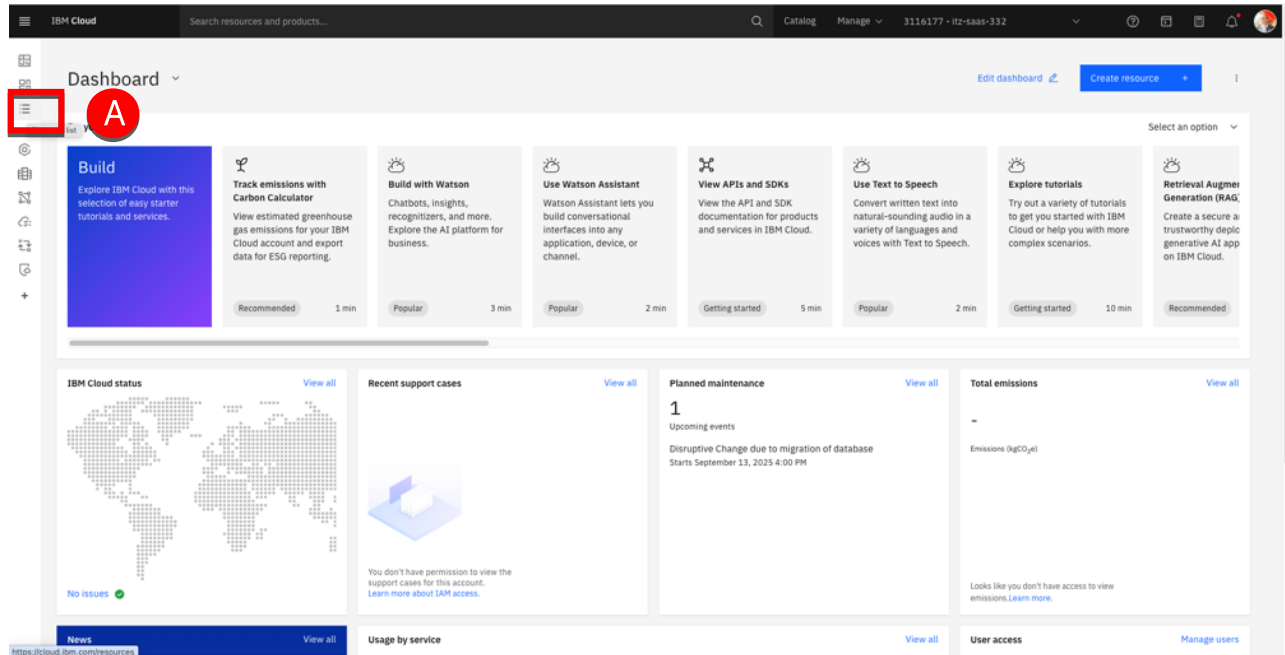
6. Make sure you are on the right cloud account as noted in item 1 in your lab credential notes. This is the IBM Cloud account named in the invitation email you received.



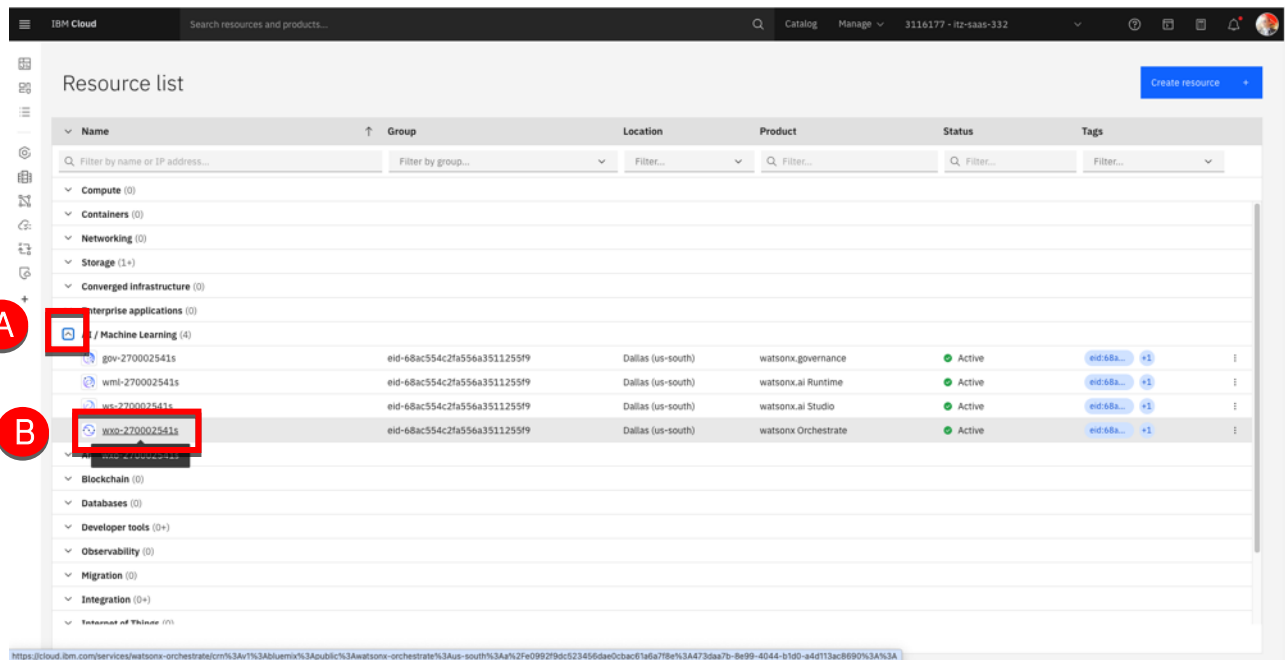
Get your instance URL

Next you will start your watsonx Orchestrate SaaS instance

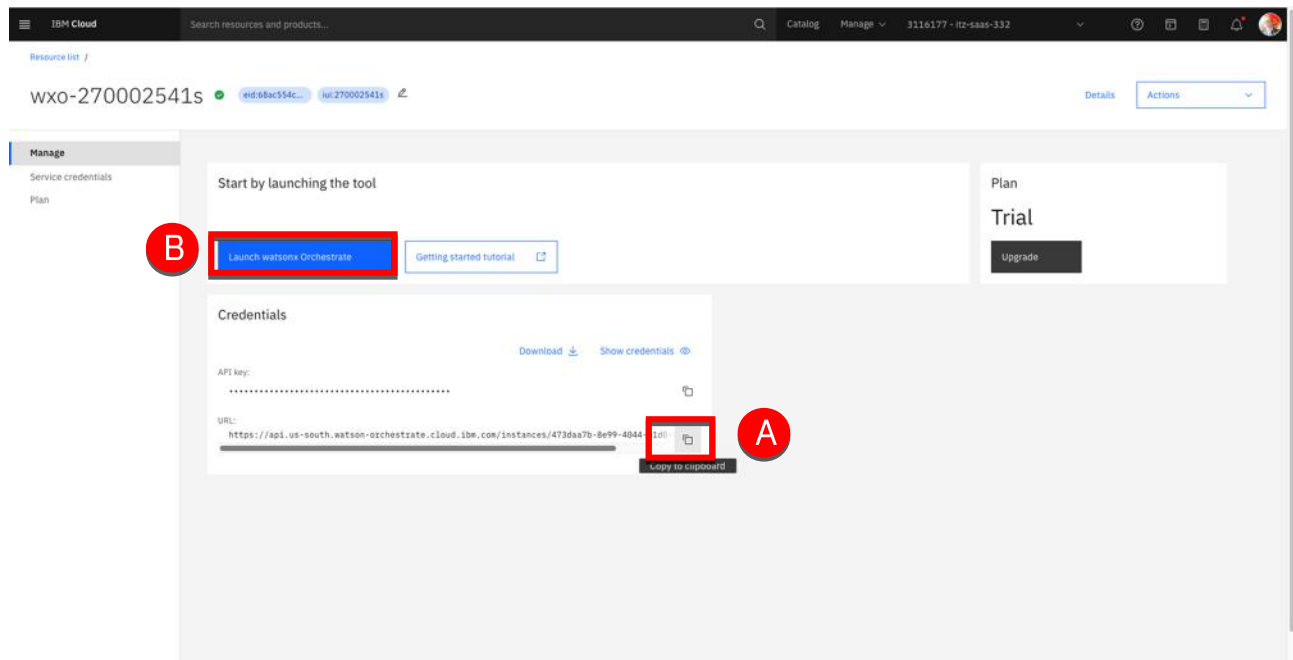
1. Click the Resource list (A).



2. Expand AI/Machine learning (A) then click watsonx Orchestrate (B).



3. Click the copy to clipboard icon next to your instance URL (A) and copy this URL in your note (Item 2). Click **Launch watsonx Orchestrate** (B).

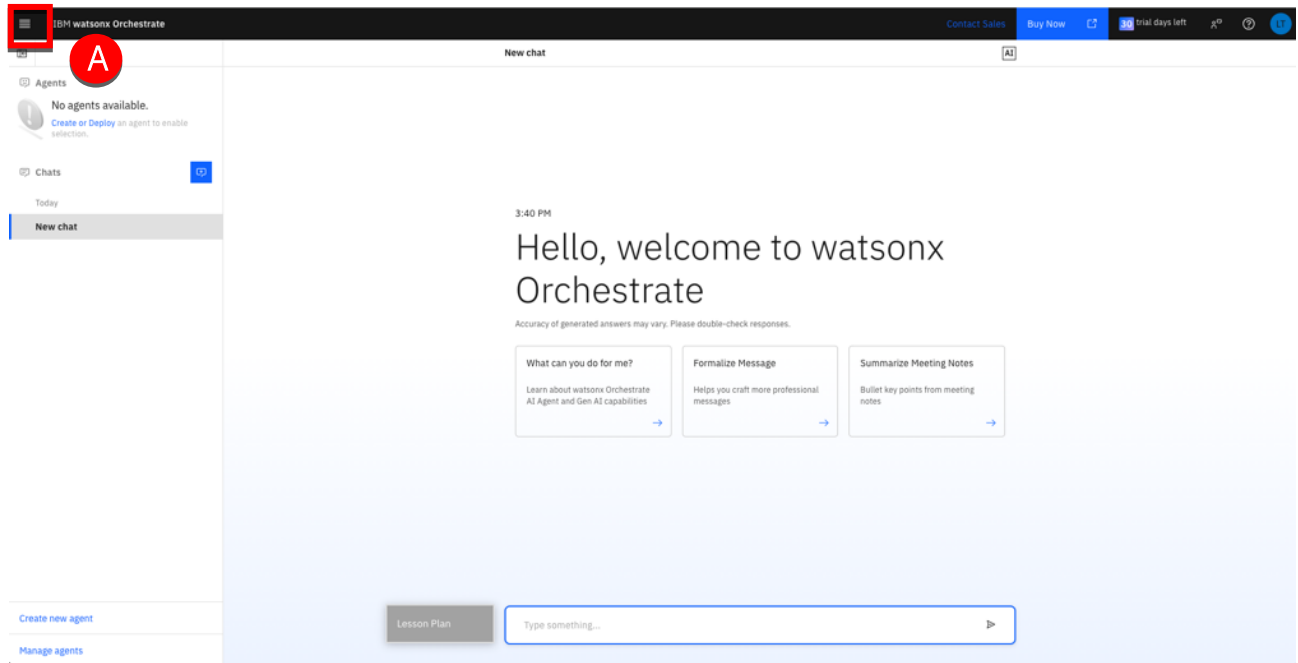


Note: Do not copy the API key from this view. This key will not work in the lab. You will have to create a new one in the following section.

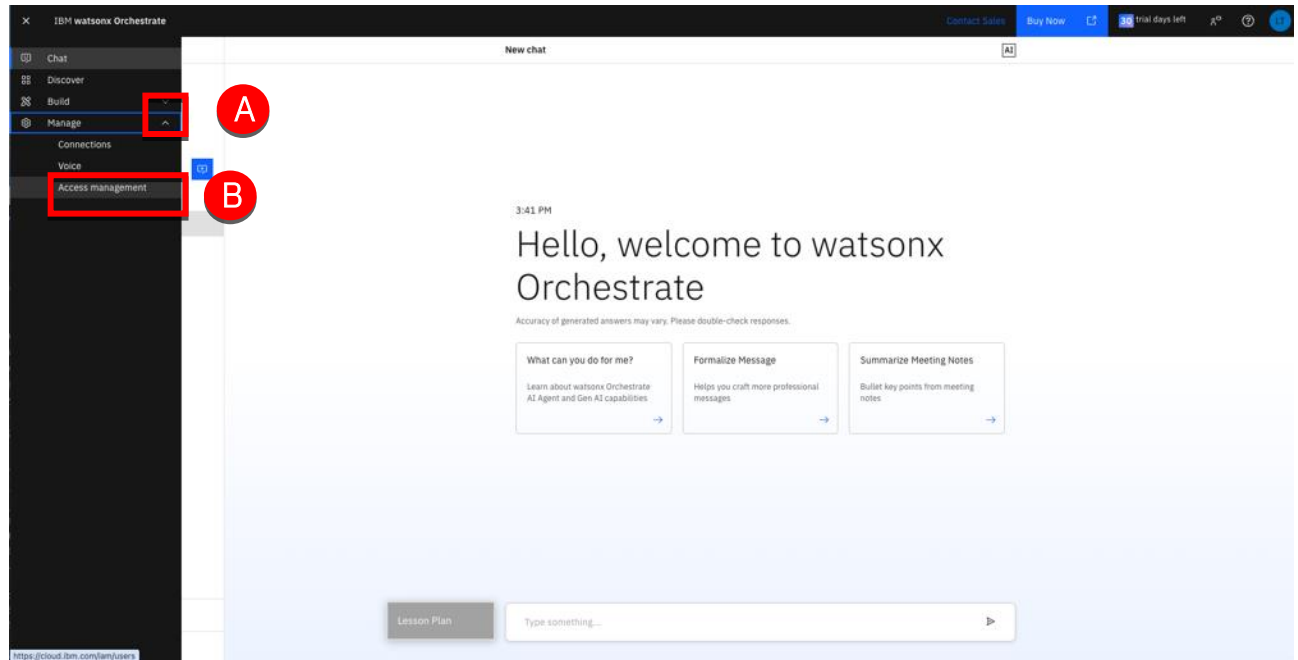
Create a new API key

Next you will create a new API key (Item 3) to access the IBM Cloud account your watsonx Orchestrate SaaS is hosted on.

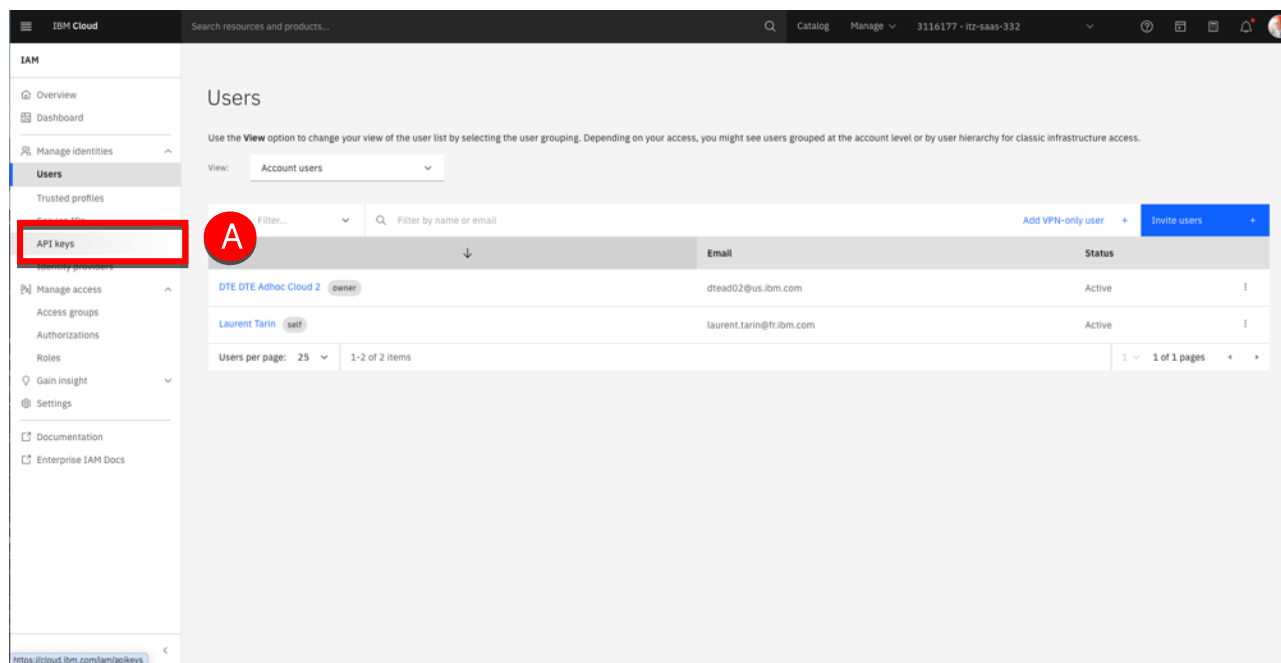
1. Click the main menu icon (A).



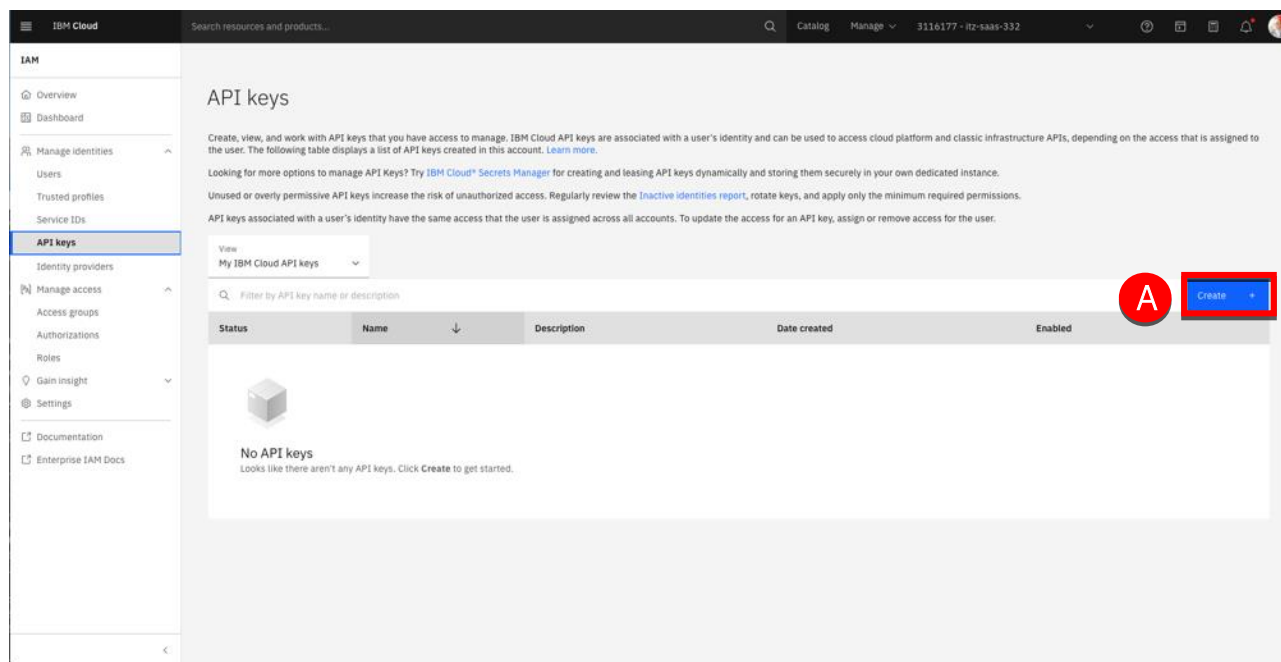
2. Expand the **Manage** menu (A) then click **Access management** (B).



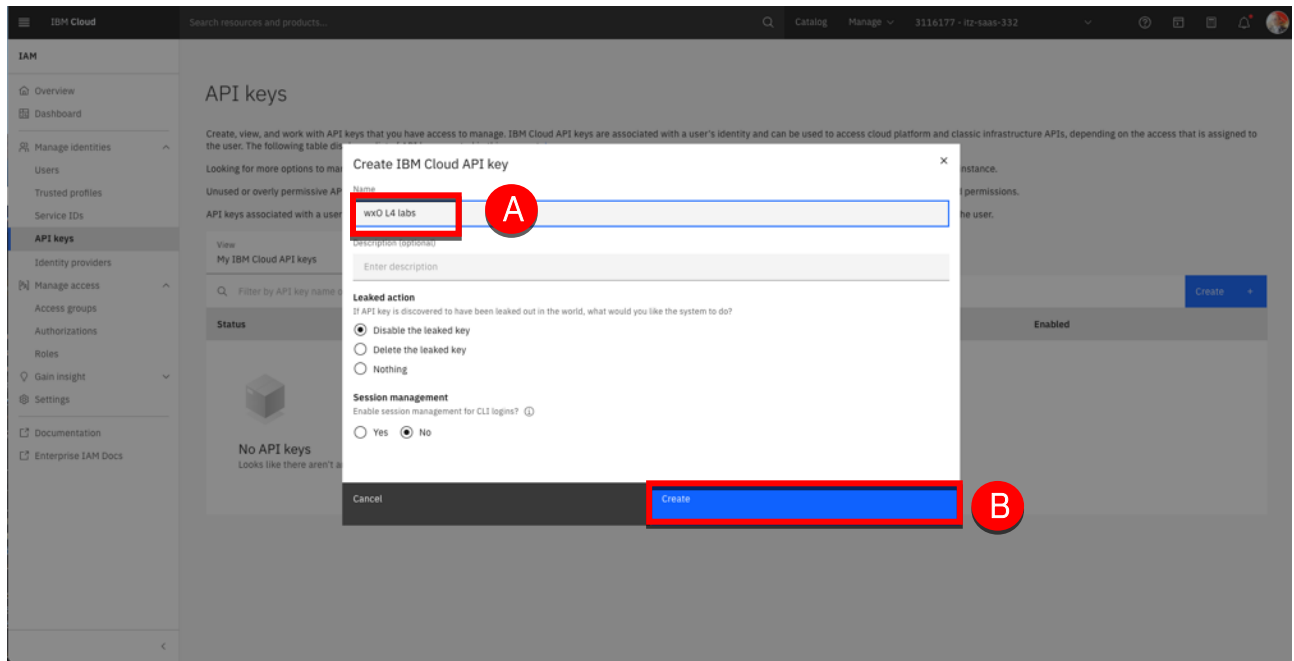
3. Click API Keys (A).



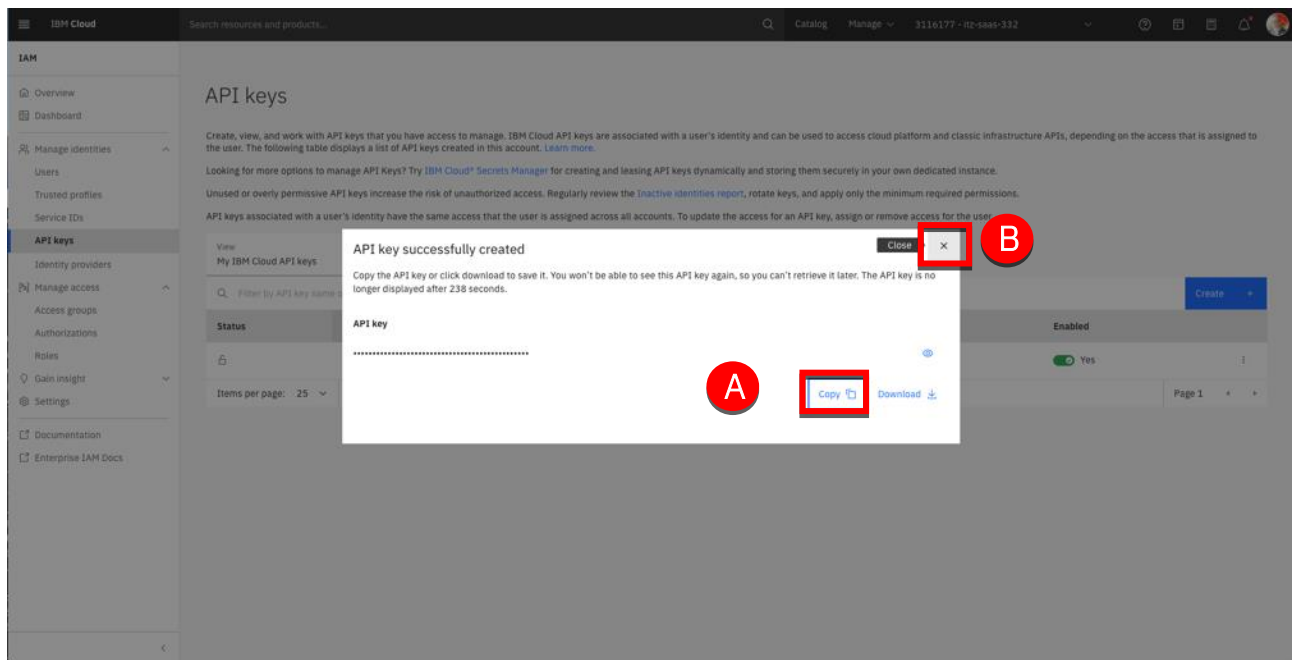
4. Click Create + (A).



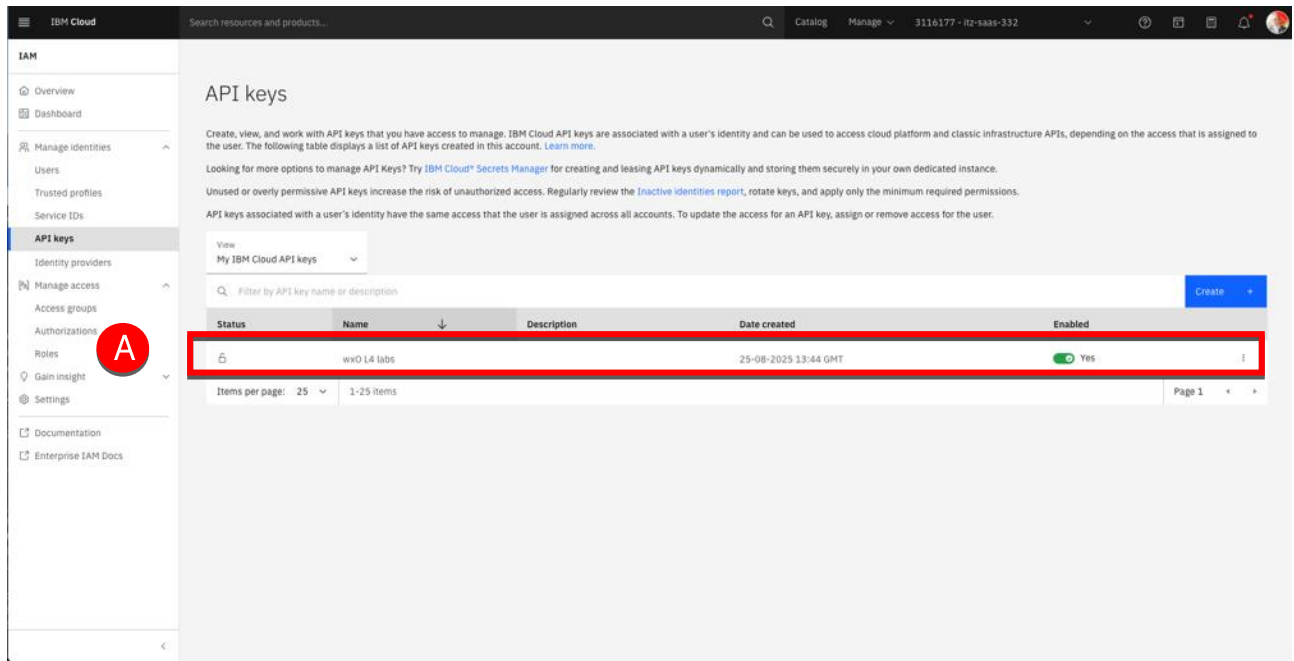
5. Enter any key name (A) then click **Create** (B).



6. Click the **Copy** icon (A) then paste the API key value in item 3 from the lab credentials table you are referencing throughout this course. Click **Close** (B).



7. Check that your key has been created (A) then close the browser tab.



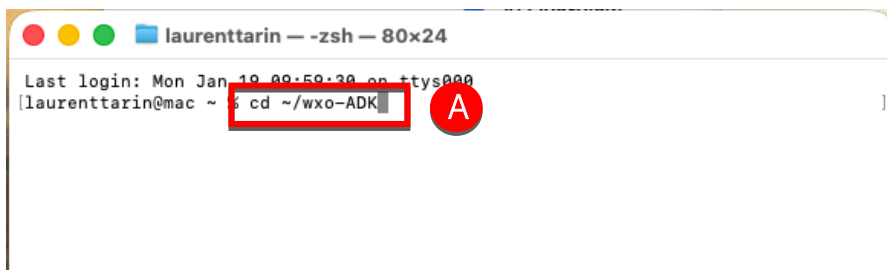
Let's now configure the watsonx Orchestrate ADK.

Install the ADK on your computer

Note: We have used a Mac to create this lab guide. Some Terminal instructions may have to be adapted should you be using a Windows terminal.

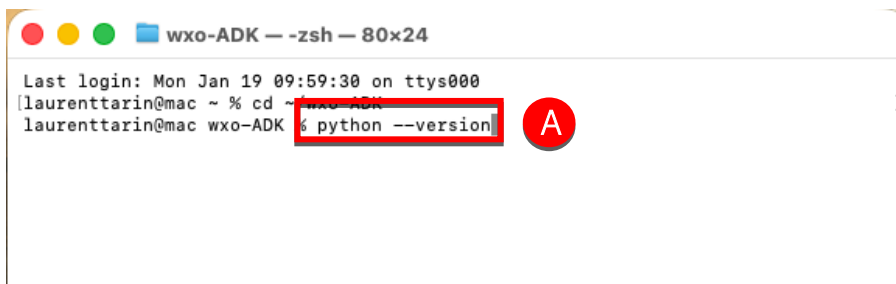
It is recommended to proceed with the ADK installation from the **wxo-ADK** folder you have created in the previous step. IT is strongly recommended to create the virtual environment (see next step) directly from this same folder. All the ADK required files are provided in this folder.

1. Open a new terminal window on your computer.
2. Type `cd ~/wxo-ADK` (A) to navigate to your local wxo-ADK folder.



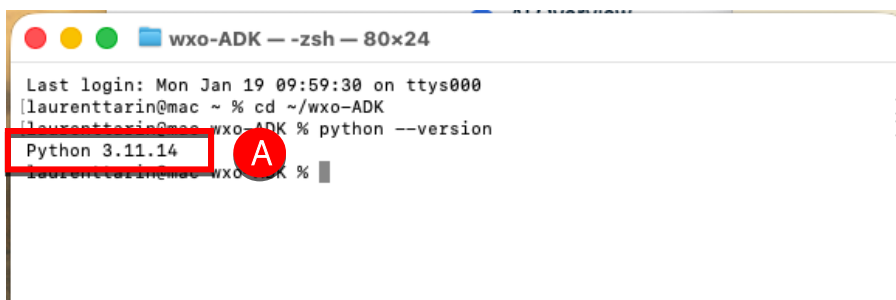
```
laurenttarin -- zsh -- 80x24
Last login: Mon Jan 19 09:59:30 on ttys000
[laurenttarin@mac ~ % cd ~/wxo-ADK]
```

3. Type `python --version` (A)



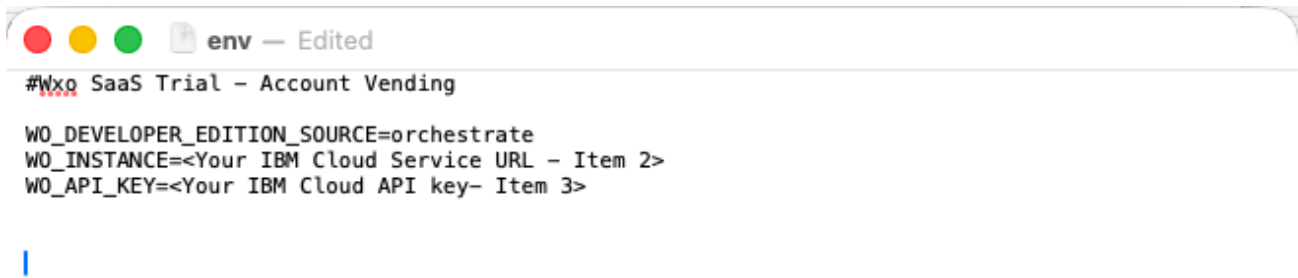
```
wxo-ADK -- zsh -- 80x24
Last login: Mon Jan 19 09:59:30 on ttys000
[laurenttarin@mac ~ % cd ~/wxo-ADK
laurenttarin@mac wxo-ADK % python --version]
```

4. Check that you are running Python 3.11 or higher (A)



```
wxo-ADK -- zsh -- 80x24
Last login: Mon Jan 19 09:59:30 on ttys000
[laurenttarin@mac ~ % cd ~/wxo-ADK
[laurenttarin@mac wxo-ADK % python --version
Python 3.11.14
laurenttarin@mac wxo-ADK % ]]
```

5. Edit the env file contained in the folder with the editor of your choice.

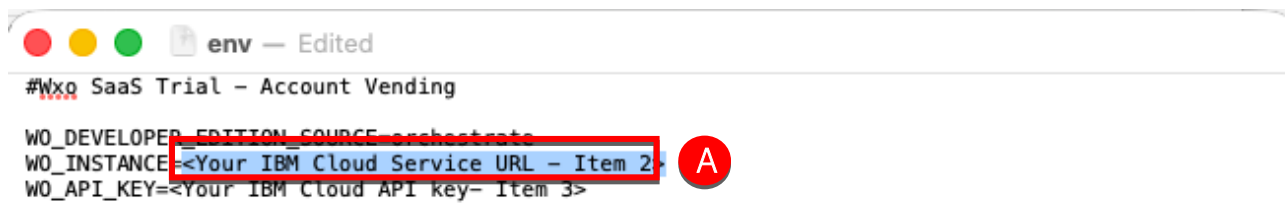


```
#Wxo SaaS Trial - Account Vending

WO_DEVELOPER_EDITION_SOURCE=orchestrate
WO_INSTANCE=<Your IBM Cloud Service URL - Item 2>
WO_API_KEY=<Your IBM Cloud API key- Item 3>
```

This file will be used to start the watsonx Orchestrate server. It provides your watsonx Orchestrate SaaS service URL and your API key to connect to your instance.

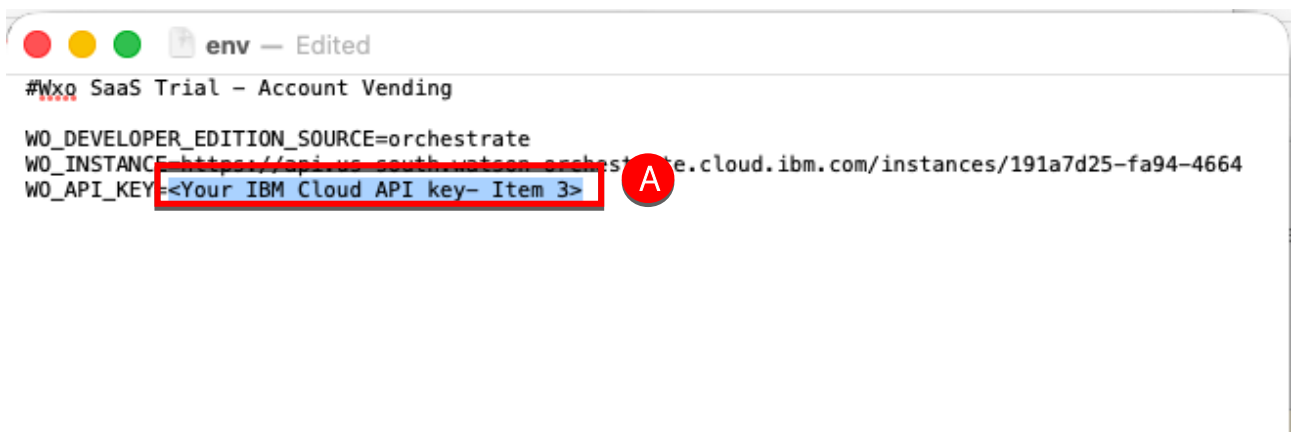
6. Edit and replace the WO_Instance with your own URL collected in item 2.



```
#Wxo SaaS Trial - Account Vending

WO_DEVELOPER_EDITION_SOURCE=orchestrate
WO_INSTANCE=<Your IBM Cloud Service URL - Item 2>
WO_API_KEY=<Your IBM Cloud API key- Item 3>
```

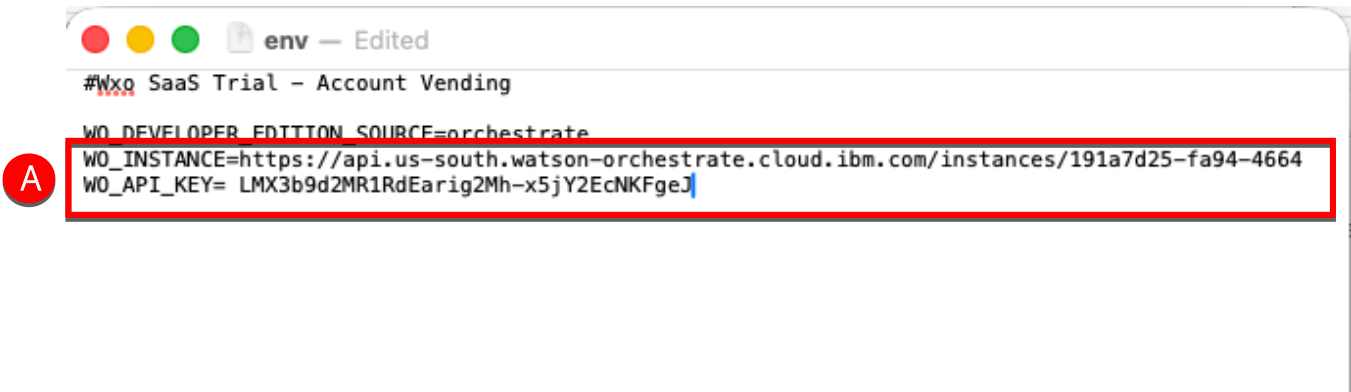
7. Edit and replace the WO_Instance with your own URL collected in item 2.



```
#Wxo SaaS Trial - Account Vending

WO_DEVELOPER_EDITION_SOURCE=orchestrate
WO_INSTANCE=https://api.us-south.watsonx-orchestrate.cloud.ibm.com/instances/191a7d25-fa94-4664
WO_API_KEY=<Your IBM Cloud API key- Item 3>
```

8. Save and exit when done (A).



```
env - Edited
#Watsonx SaaS Trial - Account Vending
WO_DEVELOPER_EDITION_SOURCE=orchestrate
WO_INSTANCE=https://api.us-south.watson-orchestrate.cloud.ibm.com/instances/191a7d25-fa94-4664
WO_API_KEY= LMX3b9d2MR1RdEarig2Mh-x5jY2EcNKFgeJ
```

Note for MS Windows users: Some MS windows users have experimented issues with their Docker configuration preventing the ADK server to start correctly. The solution was to add docker specific setup instructions in the env file:

```
DOCKER_IMAGE_PULL_LAYERS_PARALLELISM=true
DOCKER_IMAGE_PULL_PARALLEL_WORKERS_COUNT=7
USE_RANGE_REQUESTS_IN_DOCKER_IMAGE_PULLS=true
WO_VERIFY_SSL=True
IGNORE_DOCKER_LAYER_CACHING=false
```

The value used may deferrer depending on your own configuration. Refer to the [product documentation](#) for more details.

9. Proceed with the **ADK 2.6.0** set-up instructions detailed in the product documentation:
https://developer.watson-orchestrate.ibm.com/getting_started/installing.

Note: Please make sure you have reserved your watsonx Orchestrate SaaS instance before proceeding with the ADK installation on your computer. Detailed instructions to book your watsonx Orchestrate SaaS instance are provided [here](#)

If you are facing an issue with the ADK install on your computer please open an IBM Support ticket at <https://www.ibm.com/mysupport>

10. In your terminal window, type 'orchestrate --version' (A) to validate your ADK version.



11. Check that you are running the ADK 2.6.X version (A).

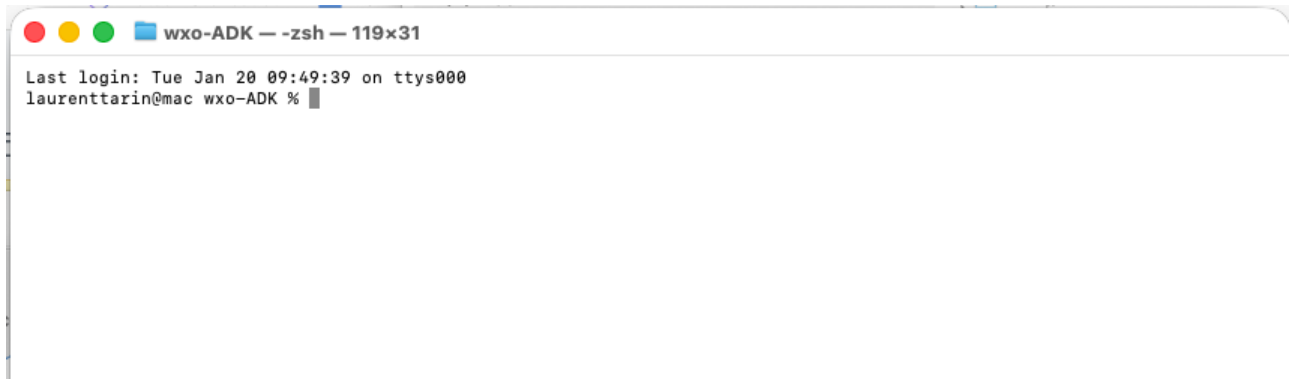
```
[ADK >orchestrate --version
ADK Version: 2.6.0
Langflow Version: 1.0.1
Developer Edition Image Tags (if not overridden in env file)
SERVER_TAG: 27-02-2026-604973067
WORKER_TAG: 27-02-2026-604973067
AI_GATEWAY_TAG: 27-01-2026-6bb3df4
MCP_GATEWAY_TAG: 20260227-develop-98-3b26d70
AGENT_GATEWAY_TAG: 05-11-2025-91a7333
DBTAG: 27-02-2026-604973067
UITAG: 25-02-2026-56fc32
CM_TAG: 27-02-2026-809cbe5
CONNECTIONS_UI_TAG: 05-11-2025-91a7333
```

Note: If you are not running the right version, type 'pip install --upgrade ibm-watsonx-orchestrate==2.6.0' to install the ADK 2.6.

Test your ADK environment

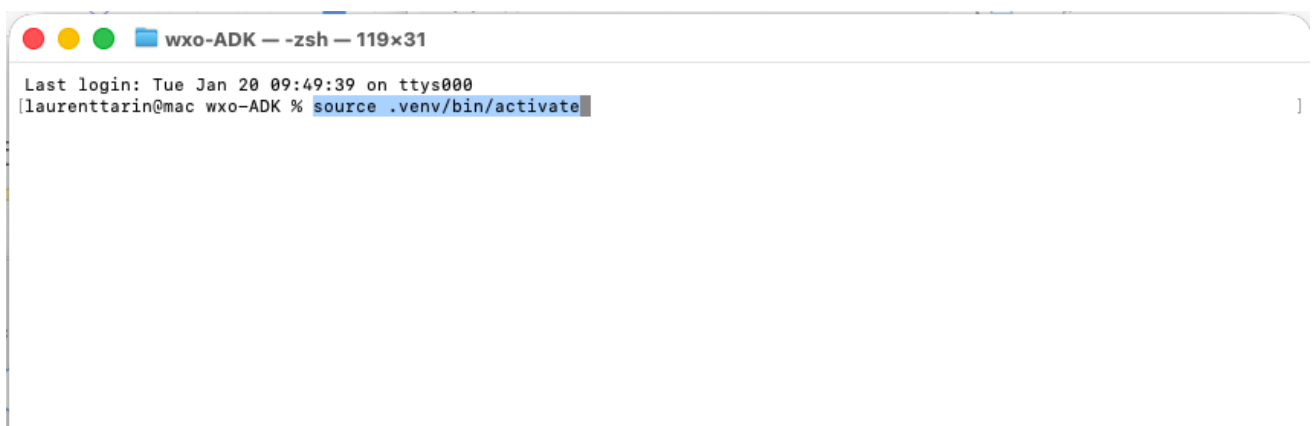
Before starting this step, you must have successfully installed the ADK on your computer following the IBM watsonx Orchestrate Documentation. Refer to the [Install](#) section for more detailed instructions you are not yet done with the install.

1. Open a Terminal window on your **wxO-ADK** folder (A).

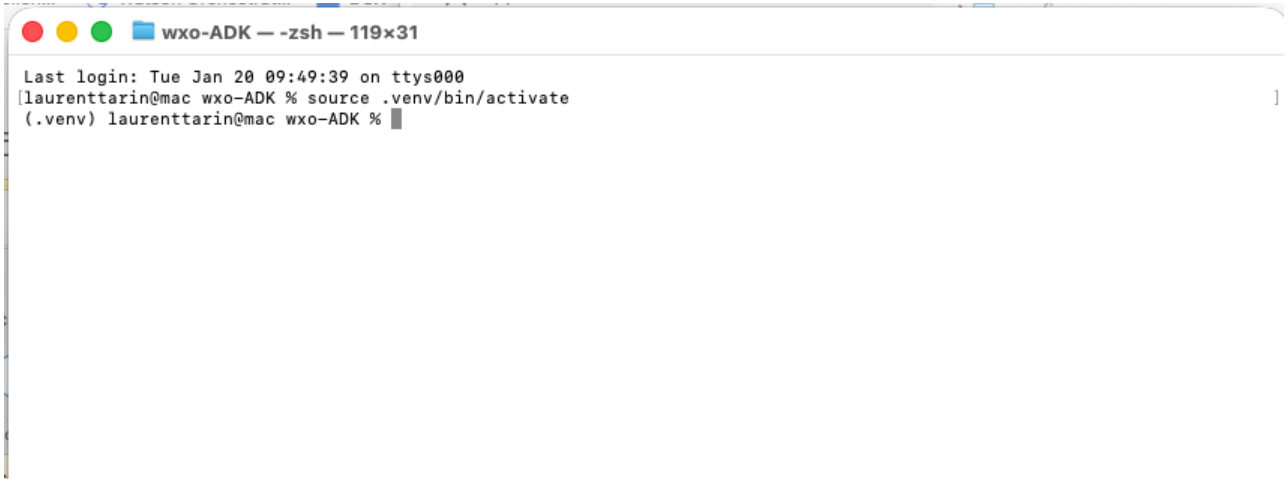


2. Enter the following command into the terminal to start your virtual environment:

```
source .venv/bin/activate
```



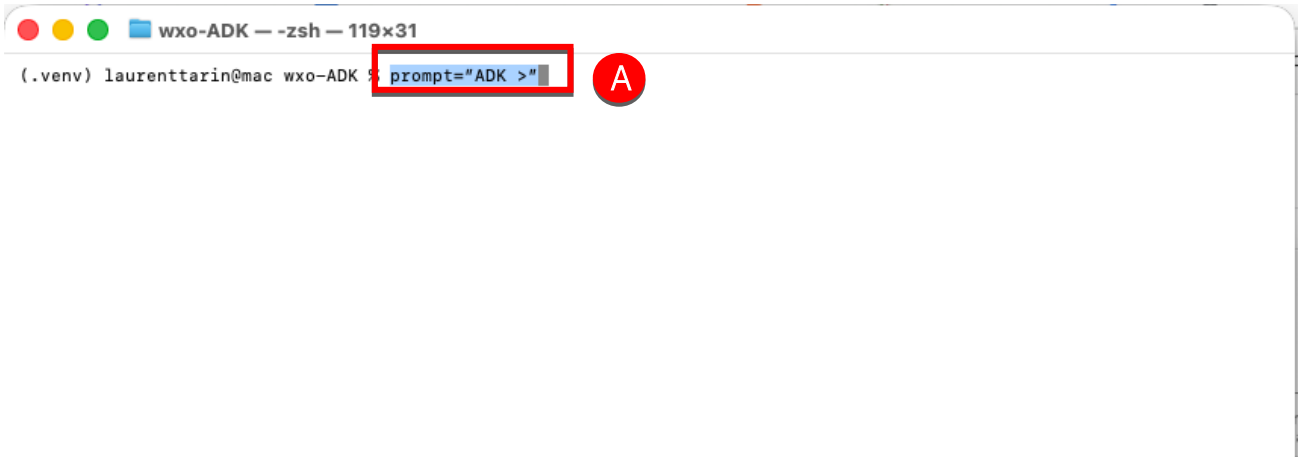
3. Check that the prompt starts with (.venv) (A) indicating that you are in your virtual environment.



```
wxo-ADK — zsh — 119x31
Last login: Tue Jan 20 09:49:39 on ttys000
[laurenttarin@mac wxo-ADK % source .venv/bin/activate
(.venv) laurenttarin@mac wxo-ADK %
```

4. Enter the following command into the terminal to improve the command visibility, then press Enter:

```
MS Windows users: function prompt{"ADK>"}
Mac users: prompt="ADK >"
```



```
wxo-ADK — zsh — 119x31
(.venv) laurenttarin@mac wxo-ADK $ prompt="ADK >"
```

A red box highlights the command `prompt="ADK >"` and a red circle with the letter 'A' is next to it.

5. Enter the following command into the terminal to start the server with your env configuration, then press Enter.

```
orchestrate server start -l -e env --with-ai-builder
```

```
wxo-ADK — laurenttarin@lima-ibm-watsonx-orchestrate: /Users/laurenttarin/wxo-ADK — zsh
ADK > orchestrate server start -l -e env --with-ai-builder
```

6. Enter a 18-character password (A) when prompted after the first server start command.

```
wxo-ADK — orchestrate server start -l -e env --with-ai-builder — 127x29

AI_GATEWAY_TAG: 27-01-2026-6bb3df4
MCP_GATEWAY_TAG: 20260227-develop-98-3b26d70
AGENT_GATEWAY_TAG: 05-11-2025-91a7333
DBTAG: 27-02-2026-604973067
UITAG: 25-02-2026-56fc32
CM_TAG: 27-02-2026-809cbe5
CONNECTIONS_UI_TAG: 25-11-2025-2129c31
TRM_TAG: 26-02-2026-398f7c3
TR_TAG: 26-02-2026-398f7c3
AR_TAG: 27-02-2026-55be48d
BUILDER_TAG: 05-03-2026-67c28fc
FLOW_RUNTIME_TAG: 09-03-2026-a527623
AGENT_ANALYTICS_TAG: 10-10-2025
JAEGER_PROXY_TAG: 09-10-2025
SOCKET_HANDLER_TAG: 06-02-2025-1bddb56
CPE_TAG: 26-11-2025-b6cb912
AGENT_ARCHITECT_TAG: 20251125-develop-60-a8ff9f8
VOICE_CONTROLLER_TAG: 12-01-2026-4df2a02
LANGFLOW_TAG: 1.7.1
WDU_TAG: 2.12.2
DOCPROC_DPS_TAG: 20260303-master-409-8d12c6d
DOCPROC_LLMSERVICE_TAG: 20260221-main-222-c633ed6
DOCPROC_CACHE_TAG: 20260224-master-194-2b3dcb3
DOCPROC_DPI_TAG: 20260304-master-440-299781bc
ETCD_TAG:
ELASTICSEARCH_TAG:
[ADK >
orchestrate server start -l -e env --with-ai-builder
Set Master Password for Local Services: ]
```

Note: This password corresponds to the password for orchestrate@ibm.com that will be used to connect Langflow. It will be displayed after each start-up of the server.

7. Wait for the images to be pulled and the server to be started.

```
wxo-ADK — limactl • orchestrate server start -l -e env --with-ai-builder — 127x29

CM_TAG: 27-02-2026-809cbe5
CONNECTIONS_UI_TAG: 25-11-2025-2129c31
TRM_TAG: 26-02-2026-398f7c3
TR_TAG: 26-02-2026-398f7c3
AR_TAG: 27-02-2026-55be48d
BUILDER_TAG: 05-03-2026-67c28fc
FLOW_RUNTIME_TAG: 09-03-2026-a527623
AGENT_ANALYTICS_TAG: 10-10-2025
JAEGER_PROXY_TAG: 09-10-2025
SOCKET_HANDLER_TAG: 06-02-2025-1bddb56
CPE_TAG: 26-11-2025-b6cb912
AGENT_ARCHITECT_TAG: 20251125-develop-68-a8ff9f8
VOICE_CONTROLLER_TAG: 12-01-2026-4df2a02
LANGFLOW_TAG: 1.7.1
WDU_TAG: 2.12.2
DOCPROC_DPS_TAG: 20260303-master-409-8d12c6d
DOCPROC_LLMSERVICE_TAG: 20260221-main-222-c633ed6
DOCPROC_CACHE_TAG: 20260224-master-194-2b3dcb3
DOCPROC_DPI_TAG: 20260304-master-440-299781bc
ETCD_TAG:
ELASTICSEARCH_TAG:
[ADK >
[ADK >orchestrate server start -l -e env --with-ai-builder
[Set Master Password for Local Services:
[INFO] - Auto-detecting local IP address for async tool callbacks...
[INFO] - Auto-configured CALLBACK_HOST_URL to: http://9.83.134.220:4321
[INFO] - Found existing VM named ibm-watsonx-orchestrate
[INFO] - VM is not running, starting...
█
```

Note: Some services may fail to start on the first attempt, and you may encounter issues connecting to the docker repository. Retry the start command in this situation, depending on the volume of requests, it can take several attempts to connect.

When startup is complete the terminal output will resemble the example below, restart can take 2-3 minutes to complete. The local server will be ready when you will see this **INFO** in the terminal screen (A):

```
wxo-ADK — -zsh — 127x29

Applying observability migration: 00_init_db.sql
Applying observability migration: 01_create_tables.sql
Applying observability migration: 02_triggers_and_functions.sql
Applying observability migration: 03_security.sql
Applying observability migration: 04_rls_policies.sql
Applying observability migration: 20250116.sql
Applying observability migration: 20251201.sql
Applying observability migration: 20260210.sql
WARN[0000] The "WXO_AWS_IAM_URL" variable is not set. Defaulting to a blank string.
WARN[0000] The "SKILLS_CATALOG_ENDPOINT" variable is not set. Defaulting to a blank string.
WARN[0000] The "TEAM_SERVER_ENDPOINT" variable is not set. Defaulting to a blank string.
WARN[0000] The "WXO_API_KEY" variable is not set. Defaulting to a blank string.
Creating mcpgateway DB
WARN[0000] The "WXO_API_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "WXO_AWS_IAM_URL" variable is not set. Defaulting to a blank string.
WARN[0000] The "TEAM_SERVER_ENDPOINT" variable is not set. Defaulting to a blank string.
WARN[0000] The "SKILLS_CATALOG_ENDPOINT" variable is not set. Defaulting to a blank string.
Creating architect DB
Applying architect migration: 20251107.sql
Applying architect migration: 20251125.sql
[INFO] - Waiting for orchestrate server to be fully initialized and ready.
[INFO] - Orchestrate services initialized successfully
[INFO] - no local tenant found. A default tenant is created
[INFO] - You can run 'orchestrate env activate local' to set your environment or 'orchestrate chat start' to start the UI service and begin chatting.
[INFO] - You can access the observability platform Langfuse at http://localhost:3000/ username: orchestrate@ibm.com, password: AZERTYUIOPQSDFGHJK
[INFO] - AI Builder feature has been enabled. You can now use AI assisted agent authoring features
```



You are now ready to work with watsonx Orchestrate ADK.

- Note:** Note the default localhost username/password that corresponds to the password you have set when starting the server for the first time.
- Note:** If you are failing to start the server due to some Docker errors, follow the following steps:
- 1- Re-launch the server start command several times. (Timeout issue)
 - 2- Generate a new API key in your SaaS Instance and try again. (API key issue)
 - 3- Delete your Techzone SaaS reservation and book a new one to be onboarded on a different cloud account (Techzone environment issue)

Additional orchestrate ADK commands:

If you need to stop, or reset your server, or stop the chat server, the following commands can be used. Note that these steps are for information only, and do not need to be performed here.

If you need to stop your server, enter the following command into the terminal, then press Enter. (Don't do this if you have just setup your environment as you will need the server running).

```
orchestrate server stop
```

To reset your server and stop it, enter the following command into the terminal, then press Enter.

```
orchestrate server reset
```

To stop your chat service, enter the following command into the terminal, then press Enter.

```
orchestrate chat stop
```

To delete all the orchestrate files and VM image Enter.

```
orchestrate server purge
```

Validate your environment

In this section you will deploy a simple agent and test it using the watsonx Orchestrate chat interface. This step is just to validate that your environment is correctly configured to perform the other labs. The configuration of the agent will not be explained in this section, it's only purpose is to validate that your environment has been configured correctly. These instructions assume your server is running.

1. Enter the following command into the terminal (A) to create an agent.

```
orchestrate agents import -f city-agent/city_agent.yaml
```

```
Skipping already applied migration: 20251023.sql
Skipping already applied migration: 20251029.sql
Skipping already applied migration: 20251101.sql
Skipping already applied migration: 20251103.sql
Skipping already applied migration: 20251104.sql
Skipping already applied migration: 20251107.sql
Skipping already applied migration: 20251110.sql
Skipping already applied migration: 20251111.sql
Skipping already applied migration: 20251112.sql
Existing wxo_observability DB found
Skipping already applied observability migration: 00_init_db.sql
Skipping already applied observability migration: 01_create_tables.sql
Skipping already applied observability migration: 02_triggers_and_functions.sql
Skipping already applied observability migration: 03_security.sql
Skipping already applied observability migration: 04_rls_policies.sql
Skipping already applied observability migration: 20250116.sql
[INFO] - Migration ran successfully.
[INFO] - Waiting for orchestrate server to be fully initialized and ready...
[INFO] - Orchestrate services initialized successfully
[INFO] - local tenant found
[INFO] - You can run 'orchestrate env activate local' to set your environment or 'orchestrate chat start' to start the
UI service and begin chatting.
[INFO] - You can access the observability platform Langfuse at http://localhost:3010, username: orchestrate@ibm.com, pa
ssword: orchestrate
ADK >orchestrate agents import -f city-agent/city_agent.yaml
```



2. Enter the following command into the terminal (A) to start the chat server.

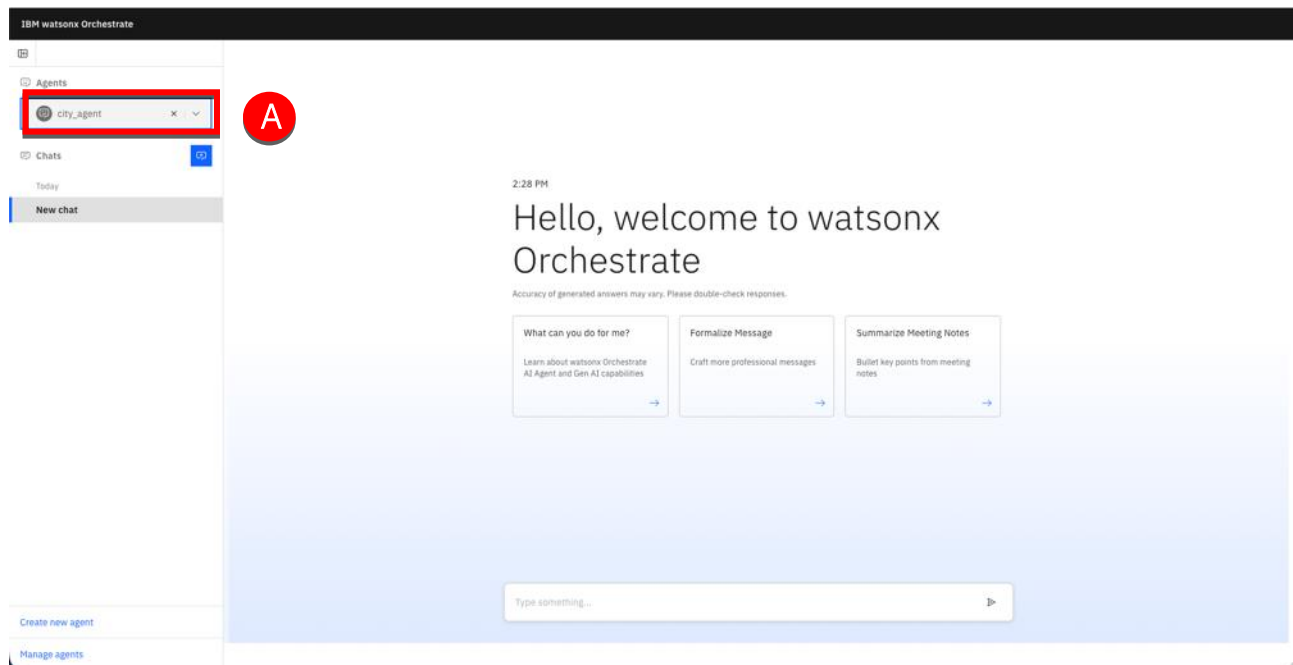
```
orchestrate chat start
```

```

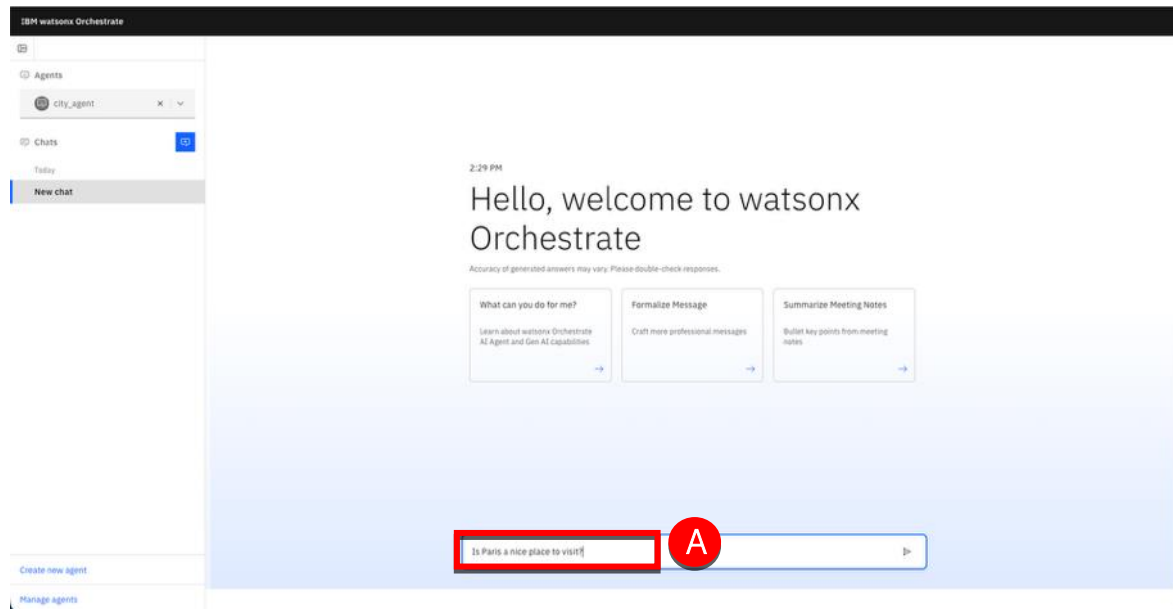
-----
Skipping already applied migration: 20251111.sql
Skipping already applied migration: 20251112.sql
Existing wxo_observability DB found
Skipping already applied observability migration: 00_init_db.sql
Skipping already applied observability migration: 01_create_tables.sql
Skipping already applied observability migration: 02_triggers_and_functions.sql
Skipping already applied observability migration: 03_security.sql
Skipping already applied observability migration: 04_rls_policies.sql
Skipping already applied observability migration: 20250116.sql
[INFO] - Migration ran successfully.
[INFO] - Waiting for orchestrate server to be fully initialized and ready...
[INFO] - Orchestrate services initialized successfully
[INFO] - local tenant found
[INFO] - You can run 'orchestrate env activate local' to set your environment or 'orchestrate chat start' to start the
UI service and begin chatting.
[INFO] - You can access the observability platform Langfuse at http://localhost:3010, username: orchestrate@ibm.com, pa
ssword: orchestrate
ADK >orchestrate agents import -f city-agent/city_agent.yaml
[INFO] - Agent 'city_agent' imported successfully
ADK >orchestrate chat start

```

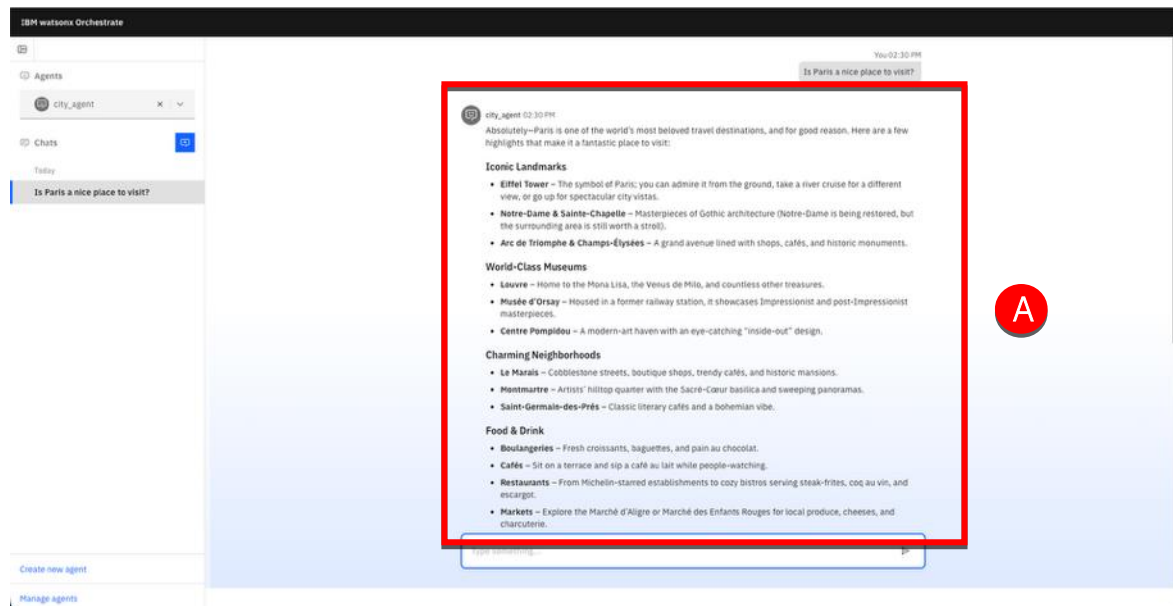
- When the browser opens, select the city-agent (A).



4. Enter “Is Paris a nice place to visit?” into the chat window (A), then press Enter.



5. Wait for the answer (A).



If the system responded with an answer, the agent is working correctly, and the system is configured correctly to work with the foundation models deployed in watsonx.ai.

Install required tools and accounts

Bruno

Bruno is a fast, open sources API client that you will use to interact with the watsonx Orchestrate APIs during this lab.

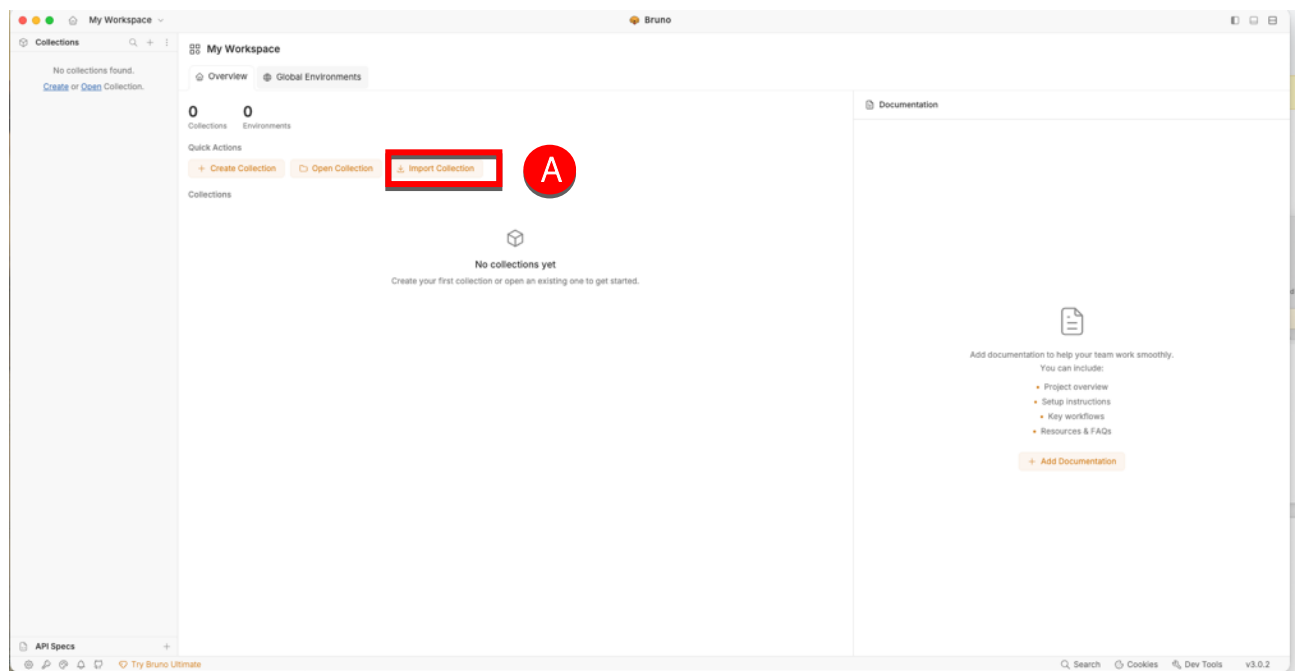
Install Bruno on your computer

To install Bruno, just visit <https://www.usebruno.com/downloads> and select the installer that corresponds to your operating system (Mac or Windows). Follow your corresponding operating system applications install procedure.

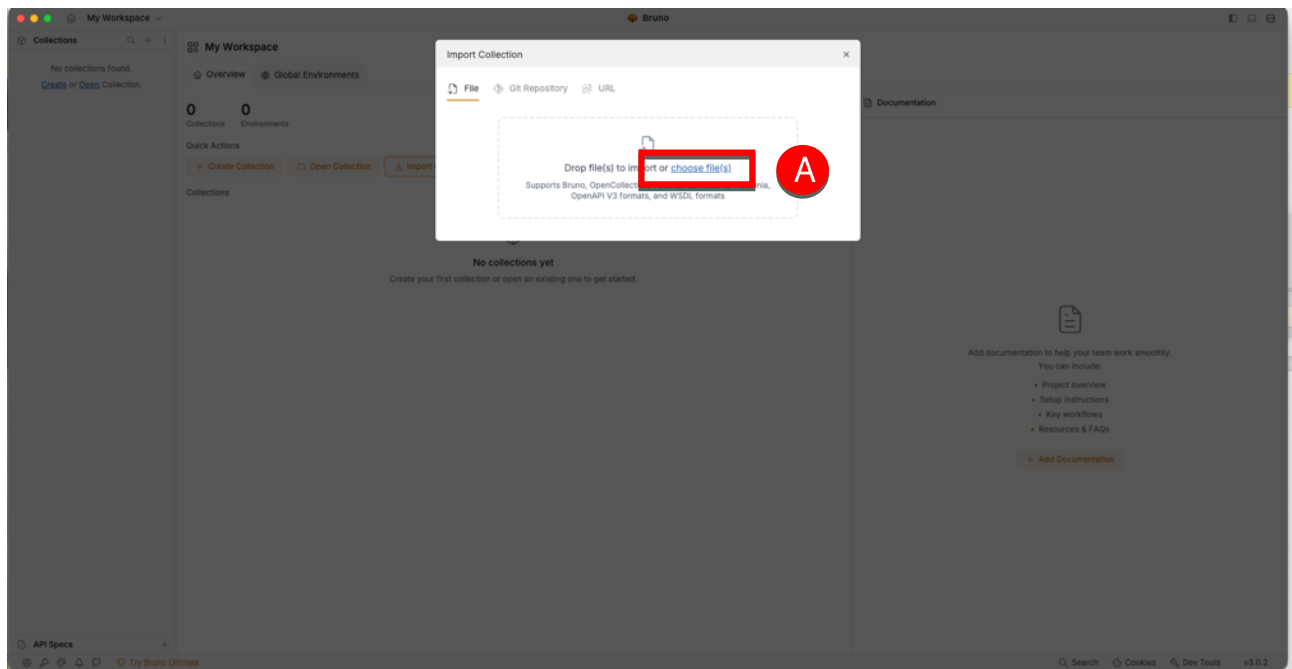
Configure Bruno

You will need to create a dedicated workspace in Bruno. The workspace definition is included in the wxO-ADK archive you have downloaded in the previous steps. These instructions will drive your through this process.

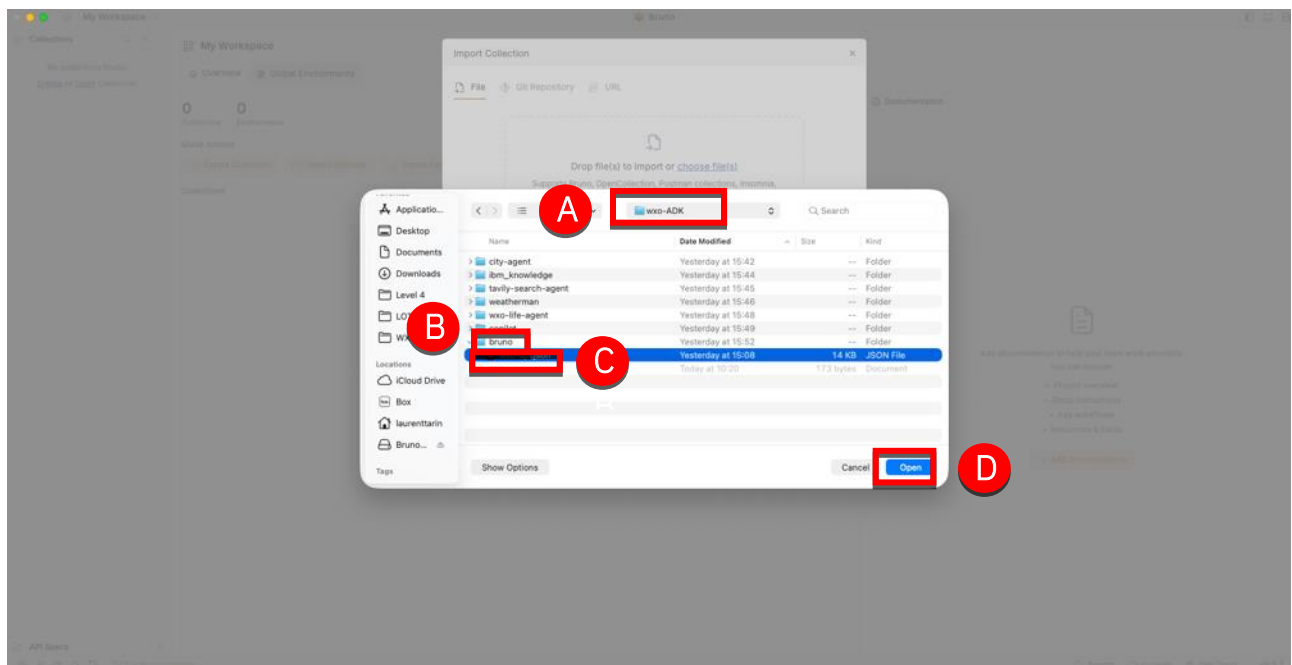
1. Start the Bruno application installed in the previous section.
2. Click **Import Collection** (A).



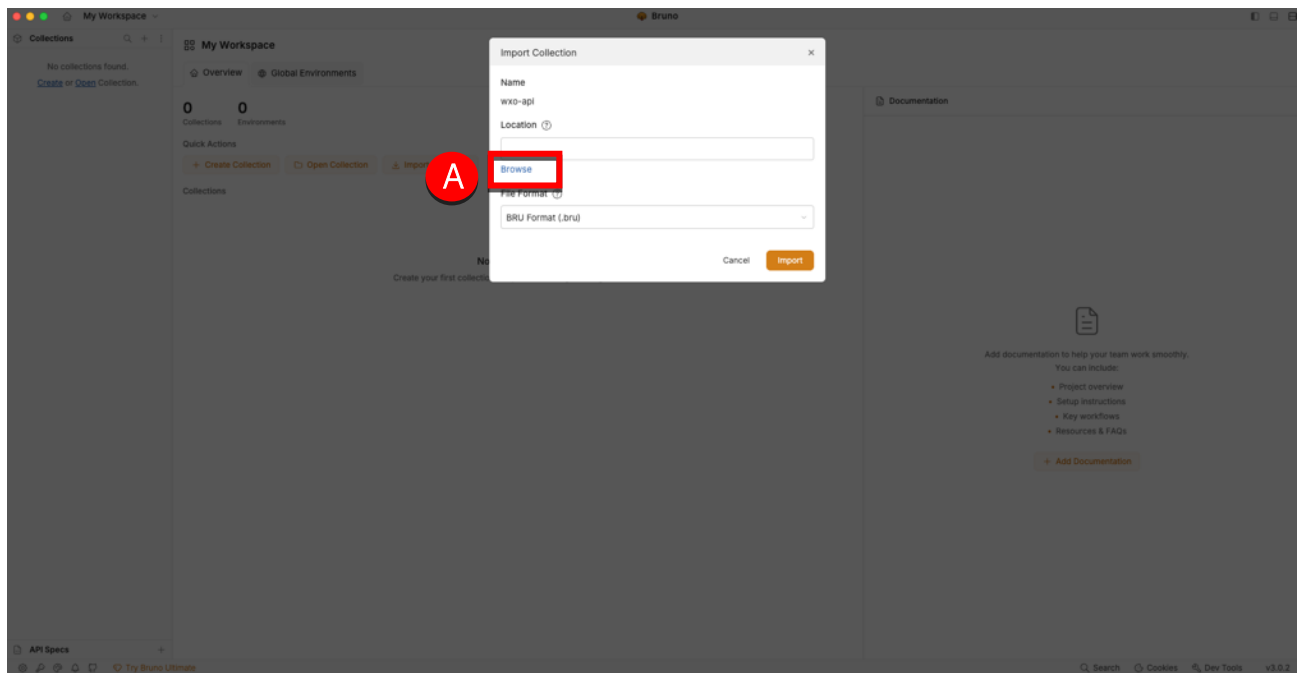
3. Click **Choose file(s)** (A).



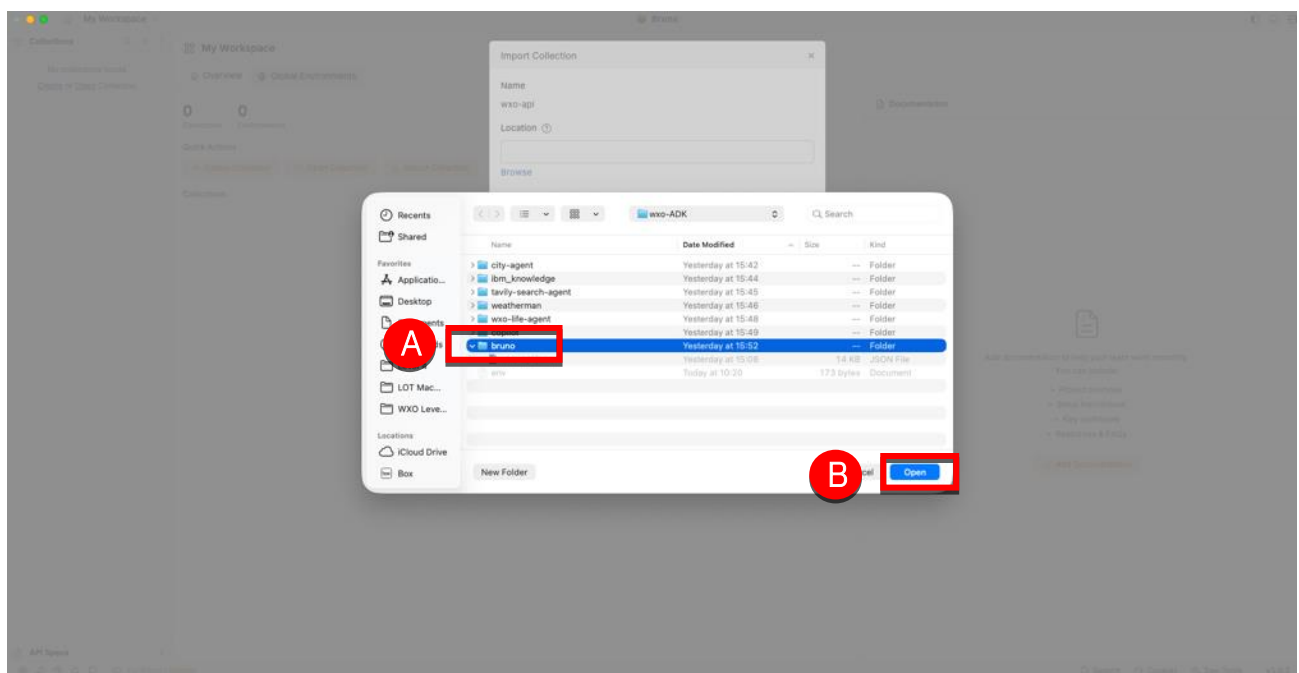
4. Go to your local **wxo-ADK** folder (A), open the **bruno** sub-folder (B) and select **wxo-api.json** file (C). Click **Open** (D).



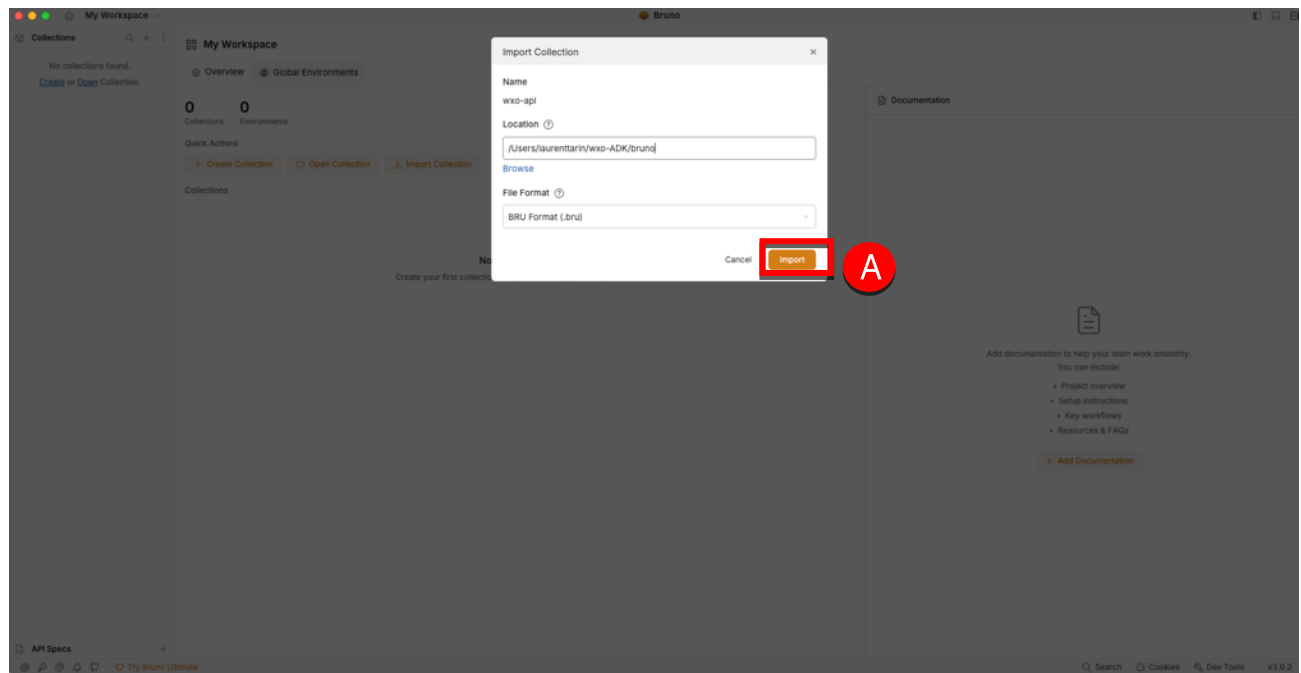
5. Click **Browse** (A) to select a file location.



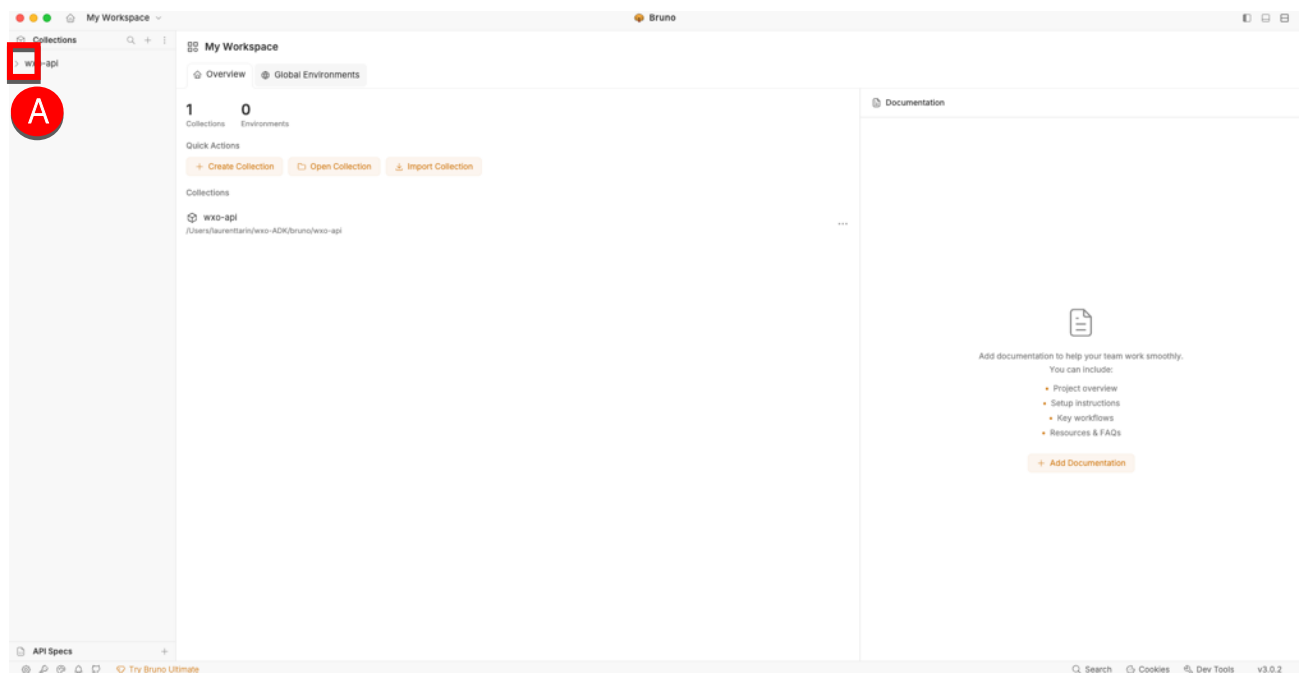
6. Select the **bruno** folder (A) and click **Open** (B).



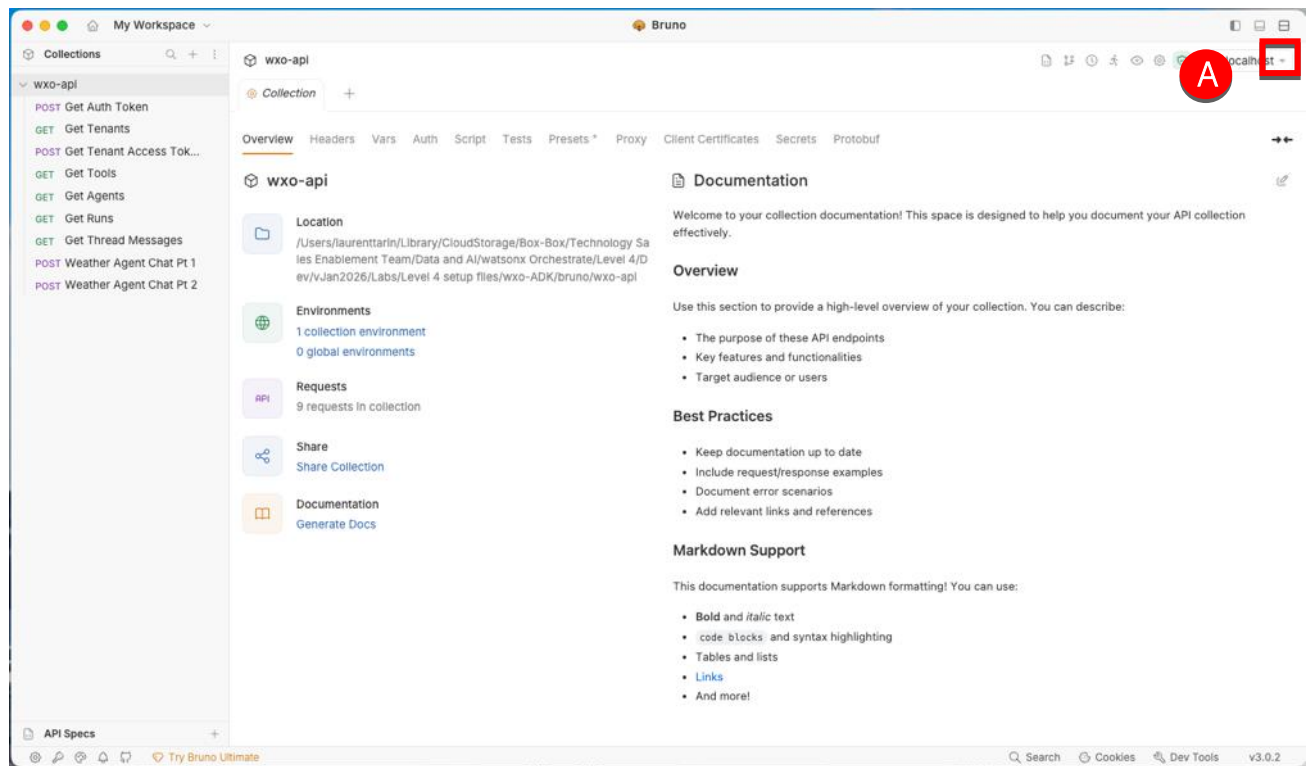
7. Click **Import** (A).



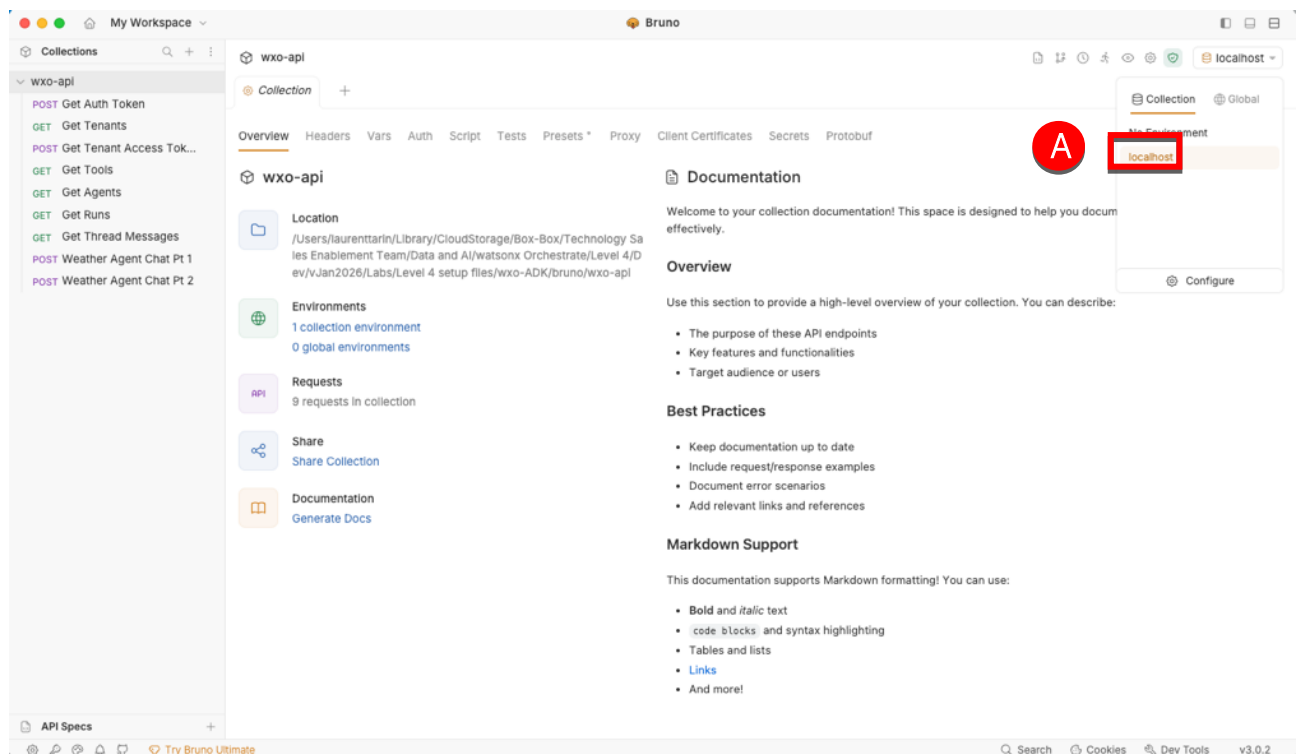
8. Expand the **wxo-api** folder (A).



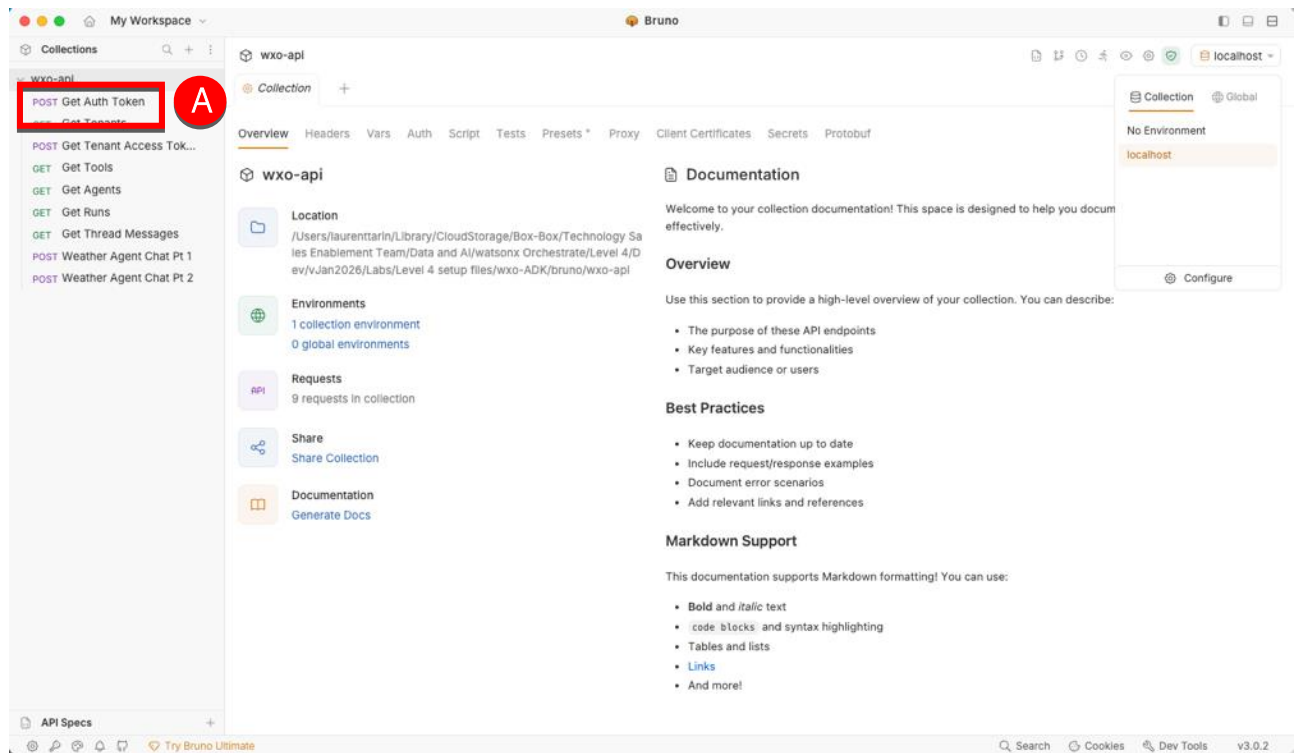
9. Expand the environment menu (A).



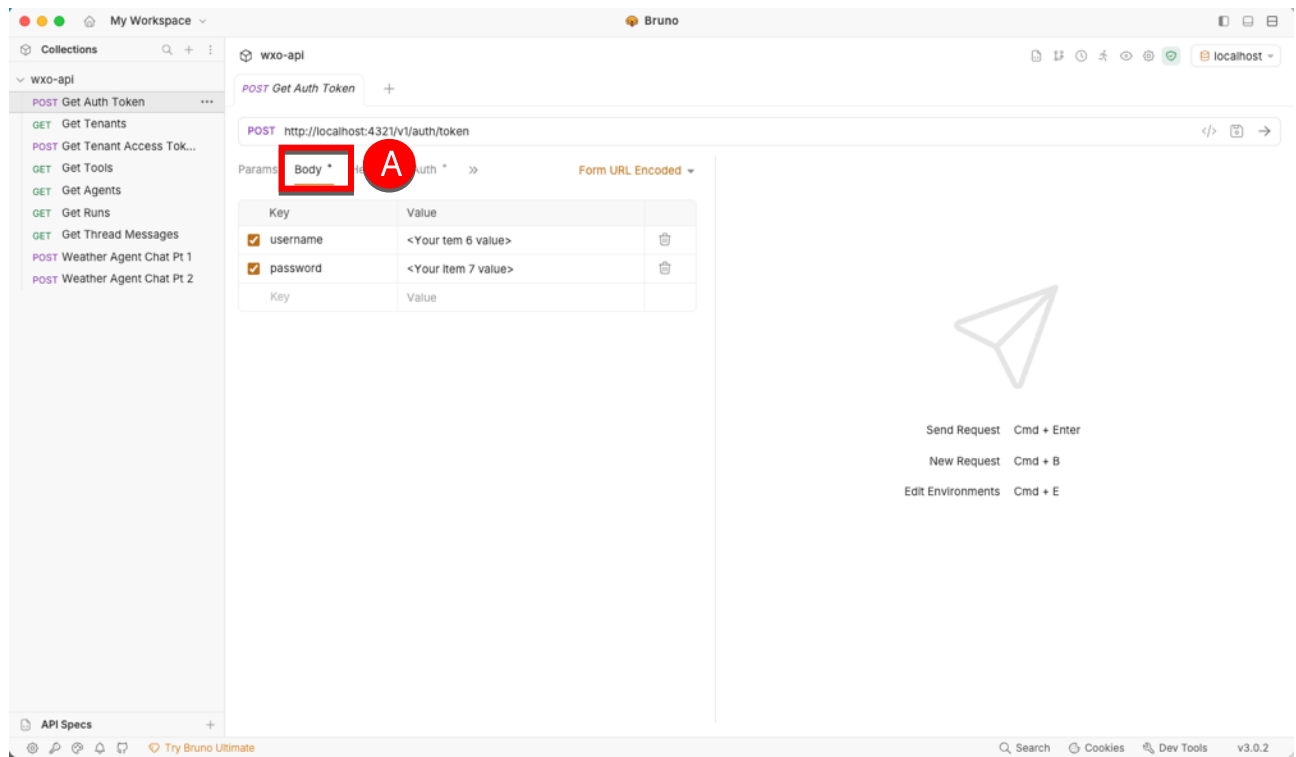
10. Select localhost (A).



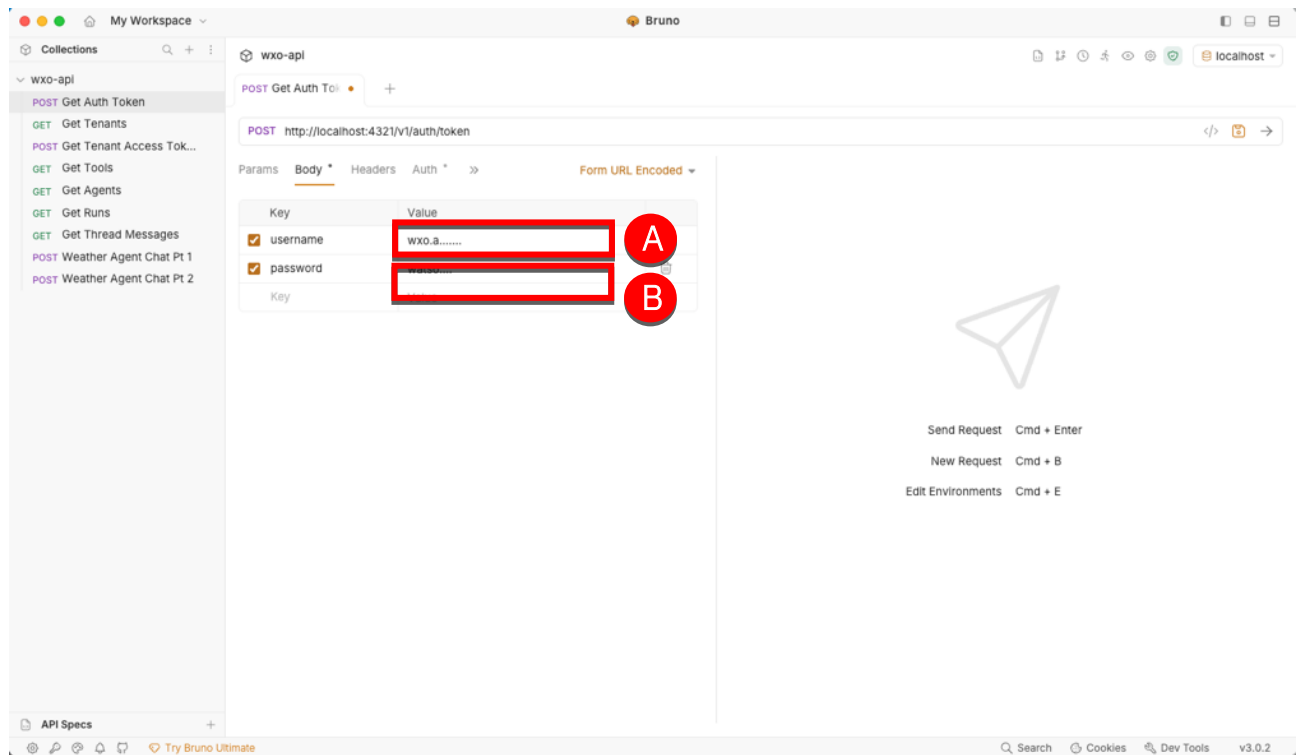
11. Click Get Auth Token (A).



12. Click the Body tab (A).

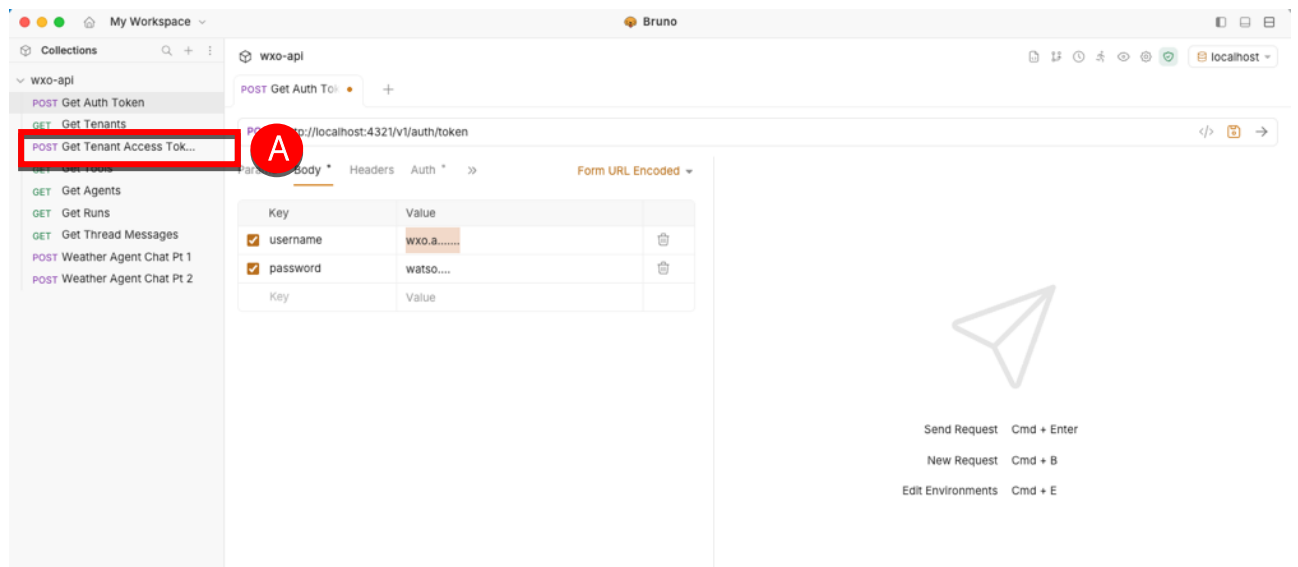


13. In the Value column, replace username with [Item 6](#) in your credentials (A) and password with [Item 7](#) in your credentials (B).

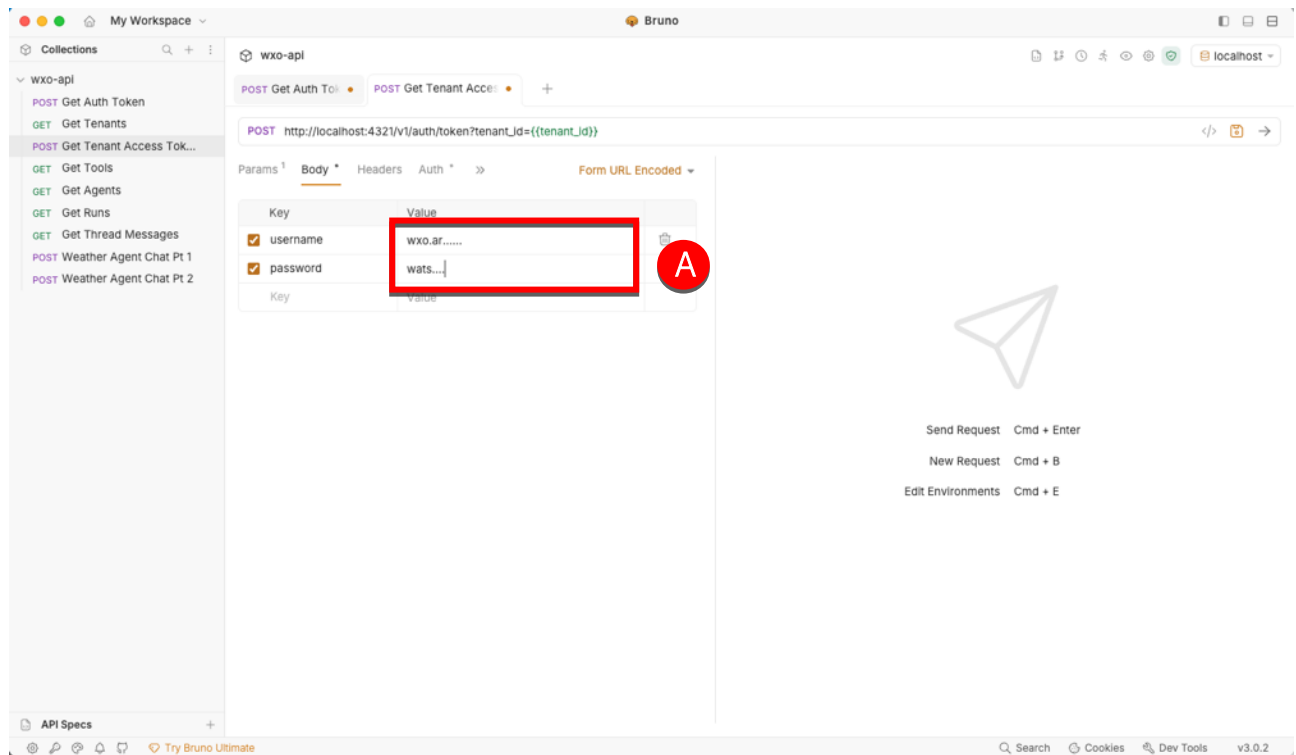


Note: the credential values have been truncated for the screenshot.

14. Click Get Tenant Access Token (A).



15. Repeat step 12-13 to set the username/password for this method (A).

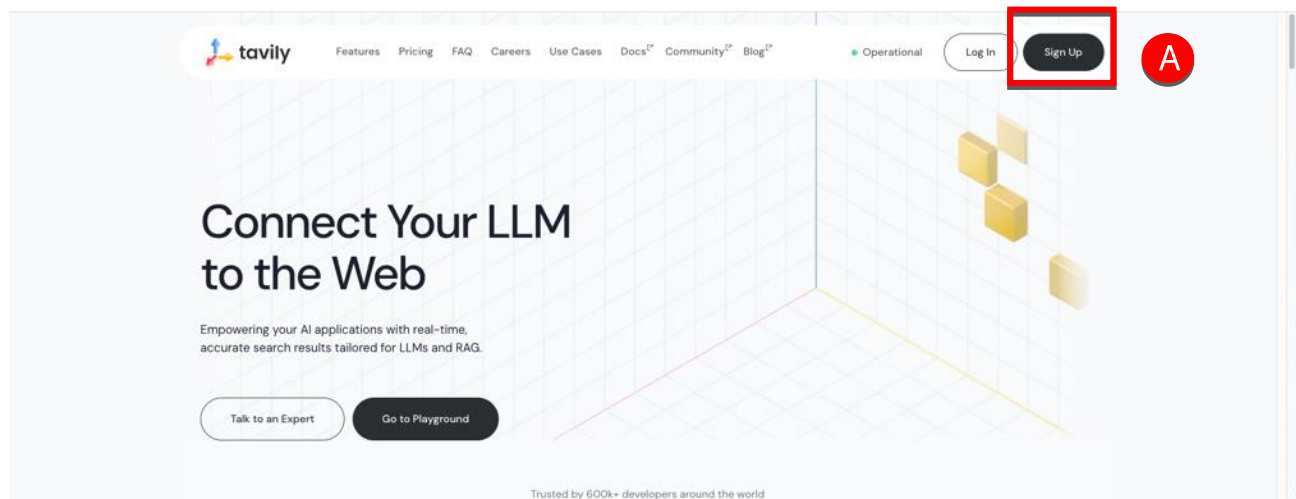


Tavily

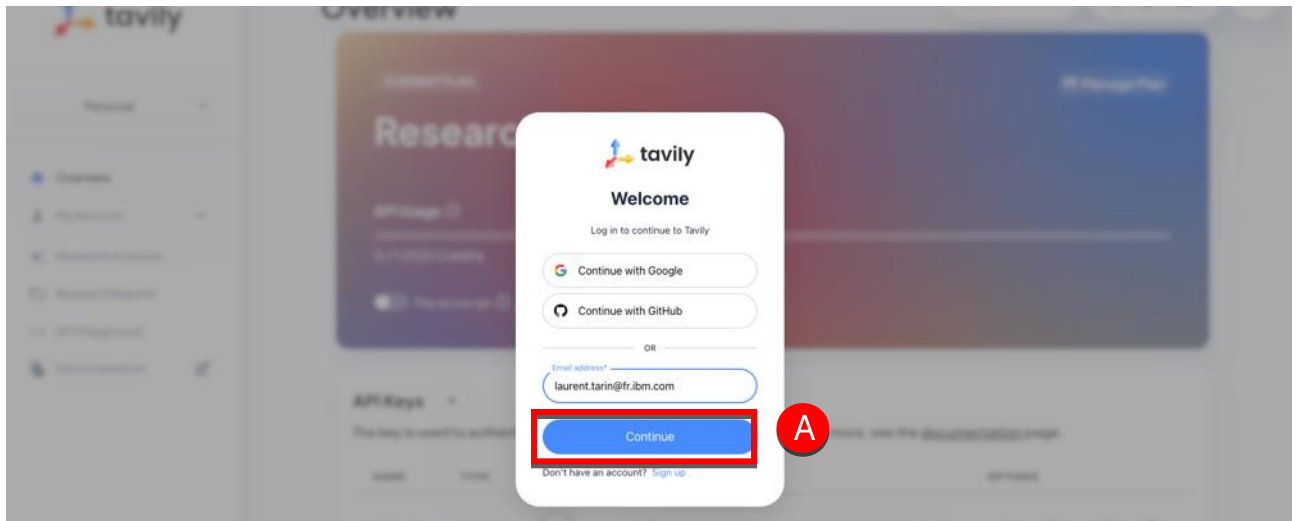
Create a free Tavily account

Tavily MCP server requires API keys. To generate this API key, you must first create a free-of-charge account on tavily.com.

1. In your web browser, go to <https://www.tavily.com/>.
2. Click **Sign-Up** (A).

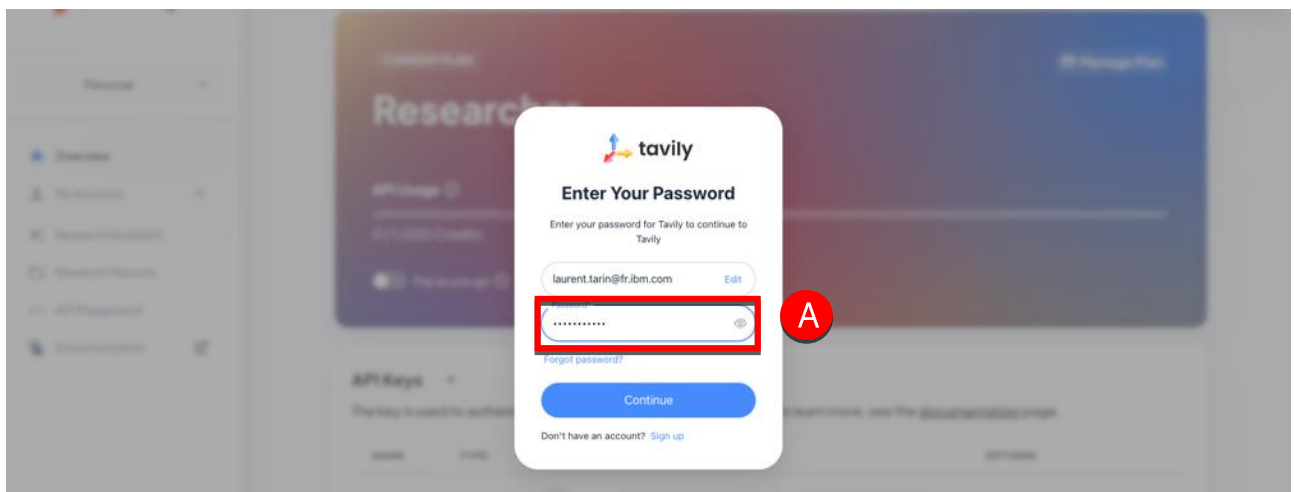


3. Enter your email address (A).

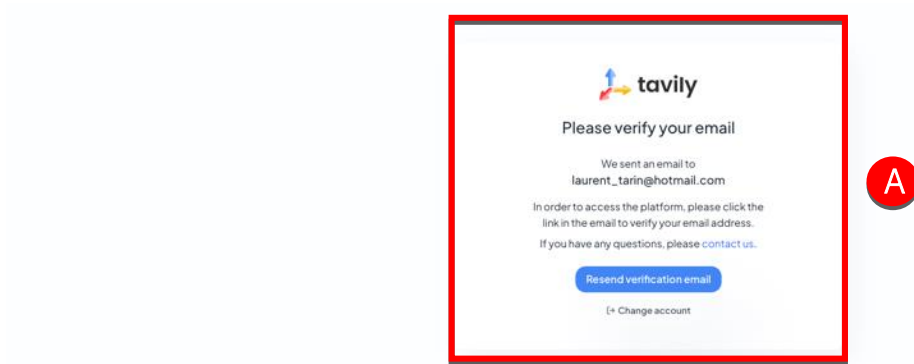


Note: If you have a google account, click Continue with google and follow the instructions to grant access from your google account.

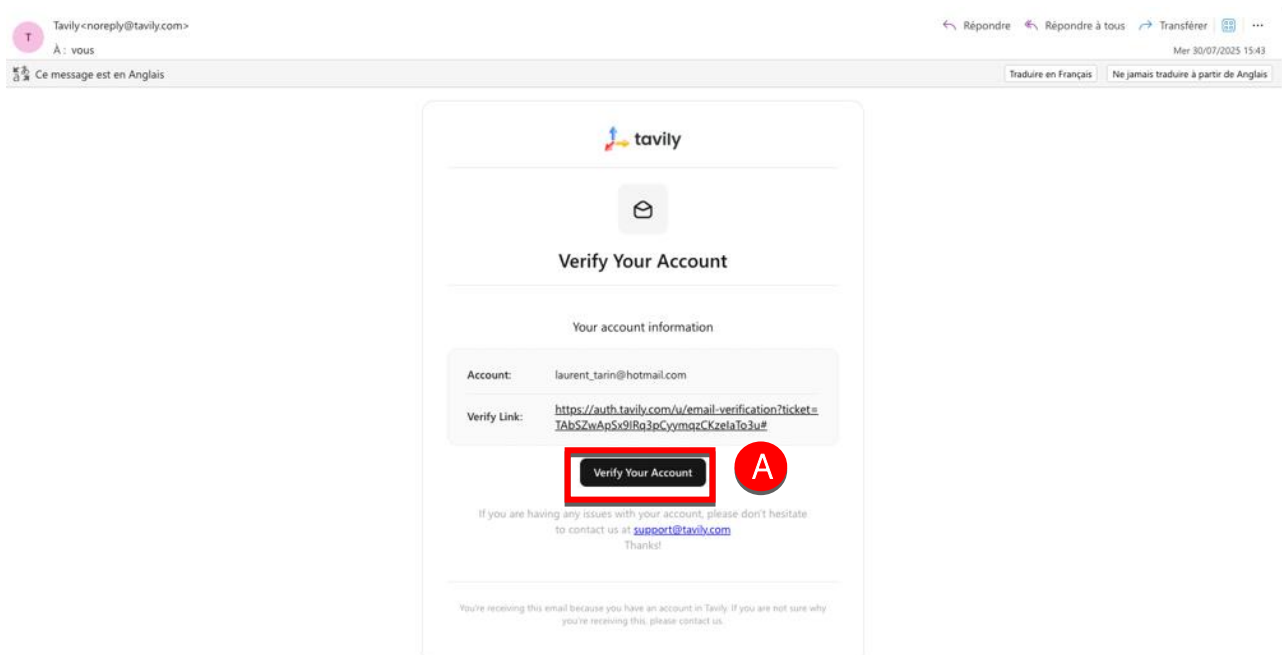
4. Enter a password (A).



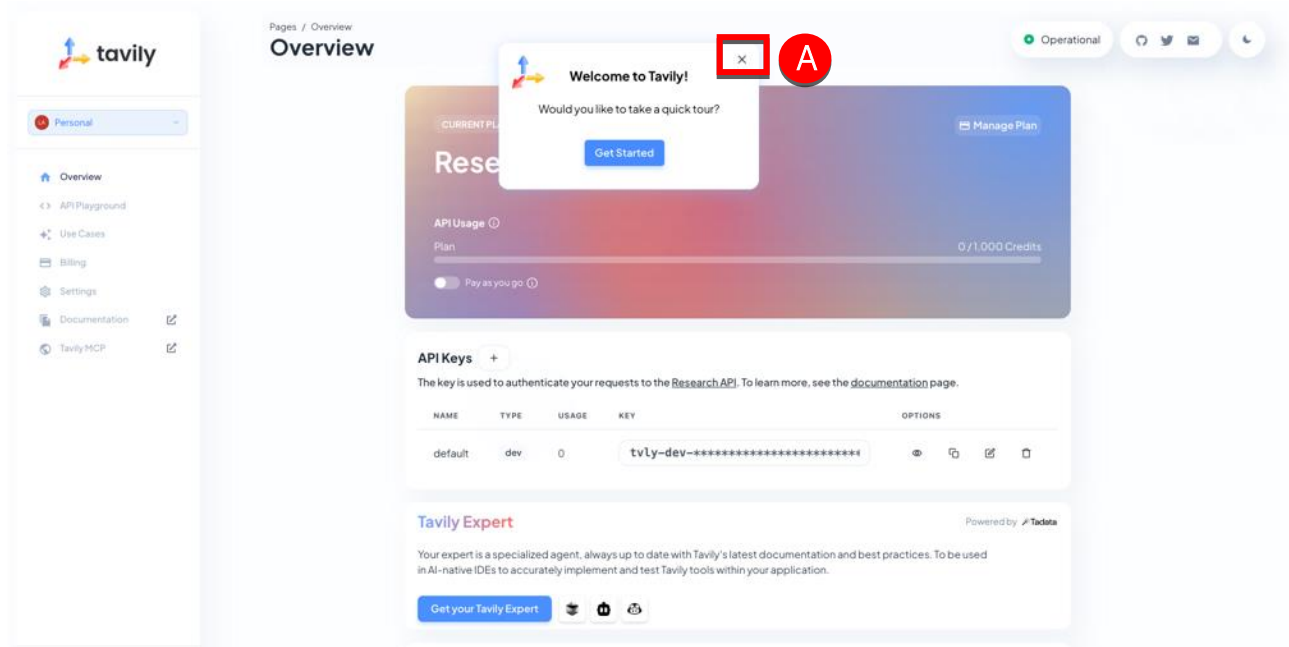
5. On this message (A), go to your corresponding email box to verify your email.



6. Open the **Verify Your Account** email received from Taviy and click **Verify Your account** (A).

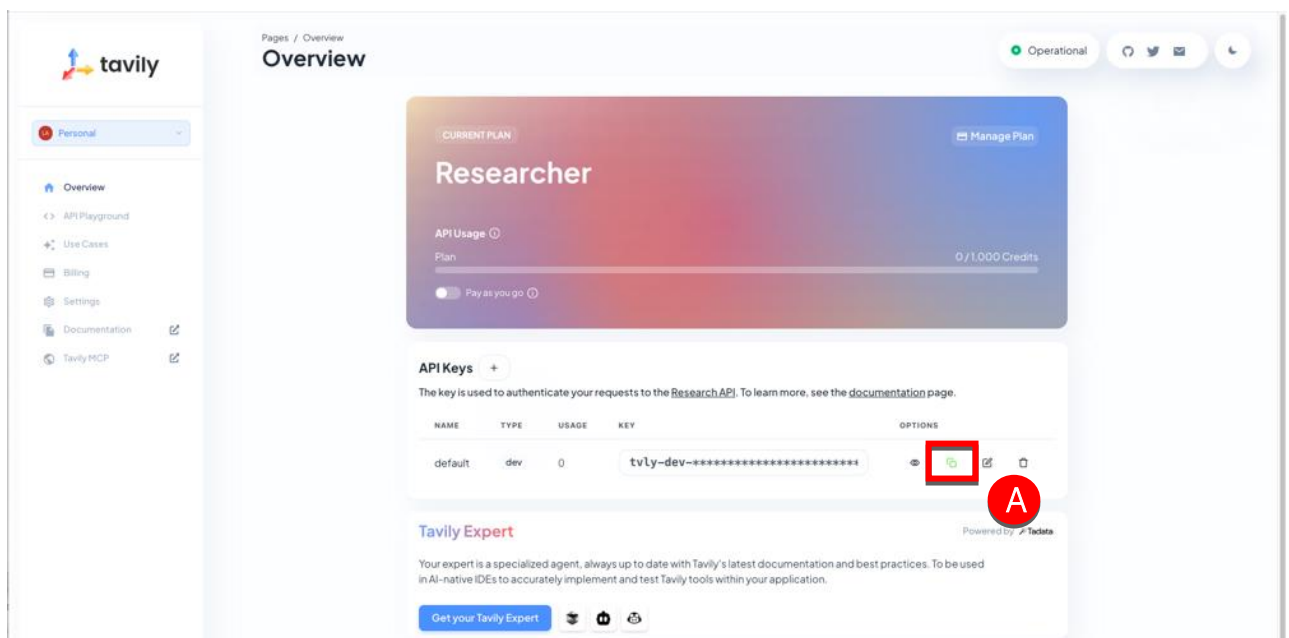


7. From the Tavily page close the **Welcome to Tavily** window (A).



Get your API key

8. A default API Key has been created for you. Click the copy to clipboard icon (A) and save the Tavily API key in your notes (Item 5).

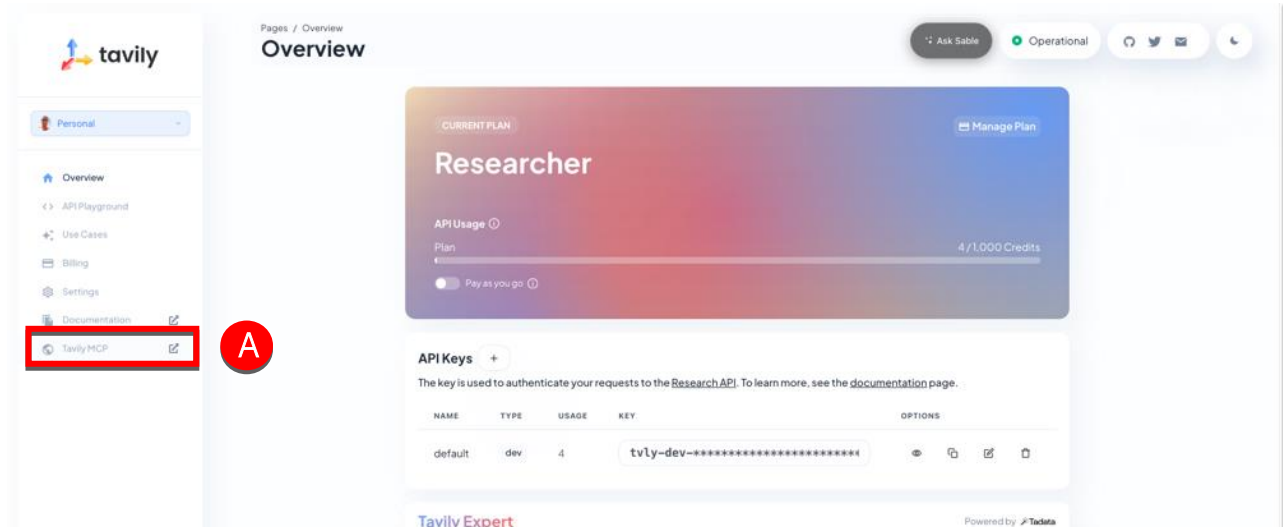


This API key you have saved will be used to configure the MCP Server in watsonx Orchestrate.

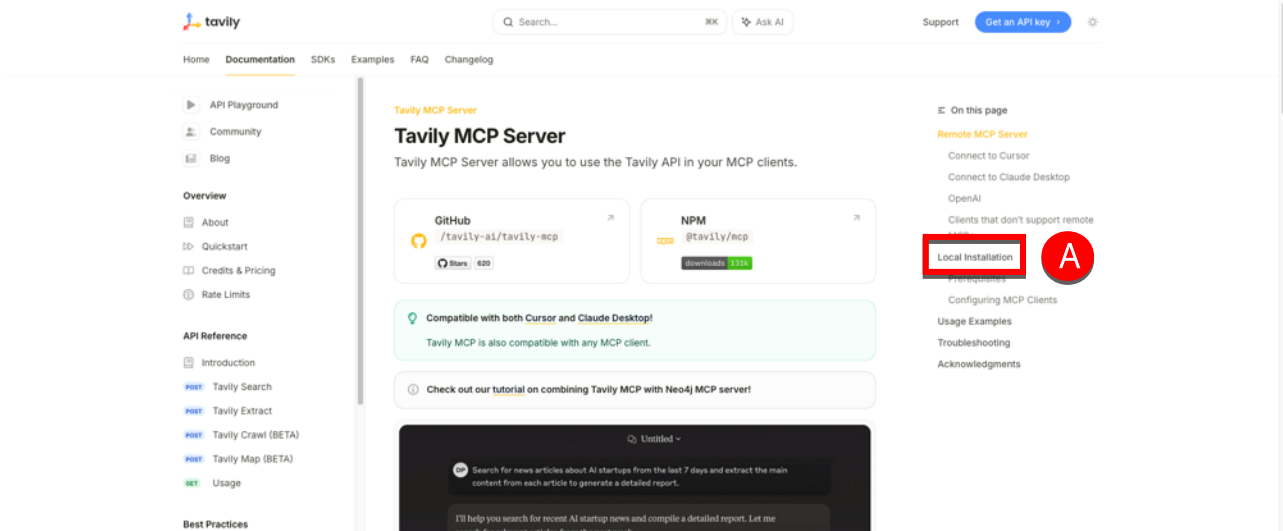
Get the MCP server installation command

Now that you have the API key, retrieve the command to install the MCP server from the MCP server documentation.

1. Click **Tavily MCP (A)**.



2. Click **Local installation (A)**.



3. Click the command line **Copy** icon (A) and copy it in your notes (Item 4). You will use this command ('`npx -y tavily -mcp@#.#.#`') later to import the MCP tools inside your watsonx Orchestrate instance.

The screenshot shows the Tavily documentation website. The left sidebar contains navigation links: Home, Documentation (selected), SDKs, Examples, FAQ, and Changelog. Under Documentation, there are links for API Playground, Community, Blog, Overview, About, Quickstart, Credits & Pricing, and Rate Limits. Under API Reference, there are links for Introduction, Tavily Search, Tavily Extract, Tavily Crawl (BETA), Tavily Map (BETA), and Usage. Under Best Practices, there are links for Best Practices for Search, Best Practices for Extract, and Best Practices for Crawl. Under Tavily MCP Server, there is a link for Tavily MCP Server (highlighted in orange). Under Partnerships, there is a link for Partnerships. The main content area is titled 'Local Installation' and includes a 'Prerequisites' section with links to 'Required Tools' and 'Git Installation (Optional)'. The 'Git Installation (Optional)' section shows the command `npx -y tavily-mcp@0.1.3` with a red box highlighting the 'Copy' icon (labeled 'A'). Below this, there is a note: 'Although you can launch a server on its own, it's not particularly helpful in isolation. Instead, you should integrate it into an MCP client.' The 'Configuring MCP Clients' section has tabs for 'Cursor' and 'Claude Desktop'. Under 'Cursor', there is a note: 'Note: Requires Cursor version 0.45.6 or higher' and a list of steps to set up the Tavily MCP server in Cursor. Under 'Claude Desktop', there is a similar list of steps. The right sidebar contains a table of contents for the page, including 'On this page', 'Remote MCP Server', 'Local Installation', 'Prerequisites', 'Configuring MCP Clients', 'Usage Examples', 'Troubleshooting', and 'Acknowledgments'.

Bump your ADK resources (optional)

In some situations (small computers / lots of running applications etc), you may face resources issues running the AI builder in your virtual environment resulting in errors. To bump your ADK resources run the following command:

1. Type `orchestrate server edit --cpus 10 --memory 24`.

```
[ADK] >orchestrate server edit --cpus 10 --memory 24
[INFO] - Editing the underlying VM settings...[INFO] - Stopping Lima VM...
[INFO] - Editing the underlying VM settings...[INFO] - Lima VM stopped.
[INFO] - Editing Lima VM...
[INFO] - Editing the underlying VM settings...[INFO] - Lima VM edited and server restarted successfully.
[INFO] - VM updated successfully.
```

Note: This new configuration should be persisted unless you run `orchestrate server purge`. Repeat this server edit command after any server purge.

Getting help

IBM employees can open an IBM Support ticket at <http://ibm.biz/support>.

Business Partners can request help at the [Partner Plus Support](#) website.

For environment provisioning issues, go to the IBM Technology Zone Help page (<https://techzone.ibm.com/help>).

Additionally, see the watsonx.ai documentation (<https://datapatform.cloud.ibm.com/docs/content/wsi/getting-started/welcome-main.html?context=wx&audience=wdp>).

Or the watsonx Orchestrate DE documentation (https://developer.watson-orchestrate.ibm.com/getting_started/wxOde_setup)

Or the watsonx Orchestrate documentation (<https://www.ibm.com/docs/en/watsonx/watson-orchestrate/base?topic=getting-started-watsonx-orchestrate>)

